

7 TIME-INDEPENDENT PERTURBATION THEORY

7.1 NONDEGENERATE PERTURBATION THEORY

7.1.1 General Formulation

Suppose we have solved the (time-independent) Schrödinger equation for some potential (say, the one-dimensional infinite square well):

$$H^0 \psi_n^0 = E_n^0 \psi_n^0, \quad (7.1)$$

obtaining a complete set of orthonormal eigenfunctions, ψ_n^0 ,

$$\langle \psi_n^0 | \psi_m^0 \rangle = \delta_{nm}, \quad (7.2)$$

and the corresponding eigenvalues E_n^0 . Now we perturb the potential slightly (say, by putting a little bump in the bottom of the well—Figure 7.1). We'd *like* to find the new eigenfunctions and eigenvalues:

$$H \psi_n = E_n \psi_n, \quad (7.3)$$

but unless we are very lucky, we're not going to be able to solve the Schrödinger equation exactly, for this more complicated potential. **Perturbation theory** is a systematic procedure for obtaining *approximate* solutions to the perturbed problem, by building on the known exact solutions to the *unperturbed* case.

To begin with we write the new Hamiltonian as the sum of two terms:

$$H = H^0 + \lambda H', \quad (7.4)$$

where H' is the perturbation (the superscript 0 always identifies the *unperturbed* quantity). For the moment we'll take λ to be a small number; later we'll crank it up to 1, and H will be the true Hamiltonian. Next we write ψ_n and E_n as power series in λ :

$$\psi_n = \psi_n^0 + \lambda \psi_n^1 + \lambda^2 \psi_n^2 + \dots; \quad (7.5)$$

$$E_n = E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2 + \dots. \quad (7.6)$$

Here E_n^1 is the **first-order correction** to the n th eigenvalue, and ψ_n^1 is the first-order correction to the n th eigenfunction; E_n^2 and ψ_n^2 are the **second-order corrections**, and so on. Plugging Equations 7.5 and 7.6 into Equation 7.3, we have:

$$\begin{aligned} & (H^0 + \lambda H') [\psi_n^0 + \lambda \psi_n^1 + \lambda^2 \psi_n^2 + \dots] \\ &= (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2 + \dots) [\psi_n^0 + \lambda \psi_n^1 + \lambda^2 \psi_n^2 + \dots], \end{aligned}$$

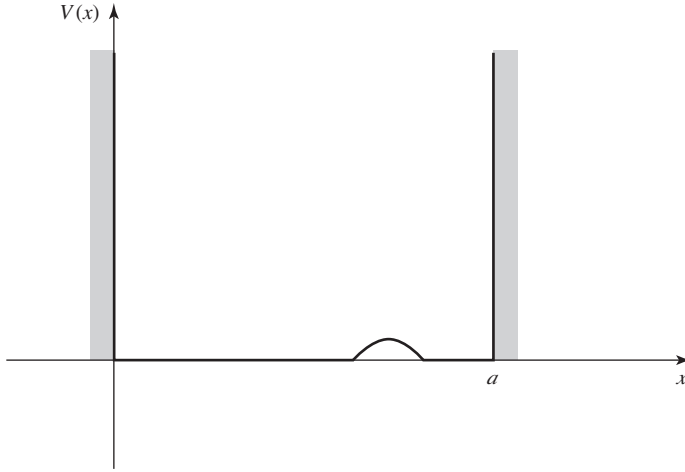


Figure 7.1: Infinite square well with small perturbation.

or (collecting like powers of λ):

$$\begin{aligned} H^0 \psi_n^0 + \lambda (H^0 \psi_n^1 + H' \psi_n^0) + \lambda^2 (H^0 \psi_n^2 + H' \psi_n^1) + \dots \\ = E_n^0 \psi_n^0 + \lambda (E_n^0 \psi_n^1 + E_n^1 \psi_n^0) + \lambda^2 (E_n^0 \psi_n^2 + E_n^1 \psi_n^1 + E_n^2 \psi_n^0) + \dots \end{aligned}$$

To lowest order¹ (λ^0) this yields $H^0 \psi_n^0 = E_n^0 \psi_n^0$, which is nothing new (Equation 7.1). To first order (λ^1),

$$H^0 \psi_n^1 + H' \psi_n^0 = E_n^0 \psi_n^1 + E_n^1 \psi_n^0. \quad (7.7)$$

To second order (λ^2),

$$H^0 \psi_n^2 + H' \psi_n^1 = E_n^0 \psi_n^2 + E_n^1 \psi_n^1 + E_n^2 \psi_n^0, \quad (7.8)$$

and so on. (I'm done with λ , now—it was just a device to keep track of the different orders—so crank it up to 1.)

7.1.2 First-Order Theory

Taking the inner product of Equation 7.7 with ψ_n^0 (that is, multiplying by $(\psi_n^0)^*$ and integrating),

$$\langle \psi_n^0 | H^0 \psi_n^1 \rangle + \langle \psi_n^0 | H' \psi_n^0 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^1 \rangle + E_n^1 \langle \psi_n^0 | \psi_n^0 \rangle.$$

But H^0 is hermitian, so

$$\langle \psi_n^0 | H^0 \psi_n^1 \rangle = \langle H^0 \psi_n^0 | \psi_n^1 \rangle = \langle E_n^0 \psi_n^0 | \psi_n^1 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^1 \rangle,$$

¹ As always (footnote 34, page 49) the uniqueness of power series expansions guarantees that the coefficients of like powers are equal.

and this cancels the first term on the right. Moreover, $\langle \psi_n^0 | \psi_n^0 \rangle = 1$, so²

$$E_n^1 = \langle \psi_n^0 | H' | \psi_n^0 \rangle. \quad (7.9)$$

This is the fundamental result of first-order perturbation theory; as a *practical* matter, it may well be the most frequently used equation in quantum mechanics. It says that the first-order correction to the energy is the *expectation value* of the perturbation, in the *unperturbed* state.

Example 7.1

The unperturbed wave functions for the infinite square well are (Equation 2.31)

$$\psi_n^0(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right).$$

Suppose we perturb the system by simply raising the “floor” of the well a constant amount V_0 (Figure 7.2). Find the first-order correction to the energies.

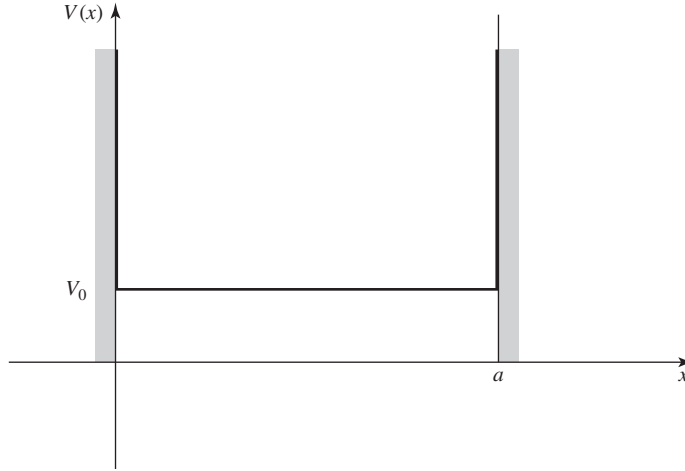


Figure 7.2: Constant perturbation over the whole well.

Solution: In this case $H' = V_0$, and the first-order correction to the energy of the n th state is

$$E_n^1 = \langle \psi_n^0 | V_0 | \psi_n^0 \rangle = V_0 \langle \psi_n^0 | \psi_n^0 \rangle = V_0.$$

The corrected energy levels, then, are $E_n \approx E_n^0 + V_0$; they are simply lifted by the amount V_0 . Of *course!* The only surprising thing is that in this case the first-order theory yields the *exact* answer. Evidently for a *constant* perturbation all the higher corrections vanish.³

² In this context it doesn't matter whether we write $\langle \psi_n^0 | H' | \psi_n^0 \rangle$ or $\langle \psi_n^0 | H' | \psi_n^0 \rangle$ (with the extra vertical bar), because we are using the wave function itself to label the state. But the latter notation is preferable, because it frees us from this convention. For instance, if we used $|n\rangle$ to denote the n th state of the harmonic oscillator (Equation 2.86), $H' |n\rangle$ makes sense, but $|H'n\rangle$ is unintelligible (operators act on *vectors/functions*, not on *numbers*).

³ Incidentally, nothing here depends on the specific nature of the infinite square well—the same holds for *any* potential, when the perturbation is a constant.

On the other hand, if the perturbation extends only half-way across the well (Figure 7.3), then

$$E_n^1 = \frac{2V_0}{a} \int_0^{a/2} \sin^2\left(\frac{n\pi}{a}x\right) dx = \frac{V_0}{2}.$$

In this case every energy level is lifted by $V_0/2$. That's not the *exact* result, presumably, but it does seem reasonable, as a first-order approximation.

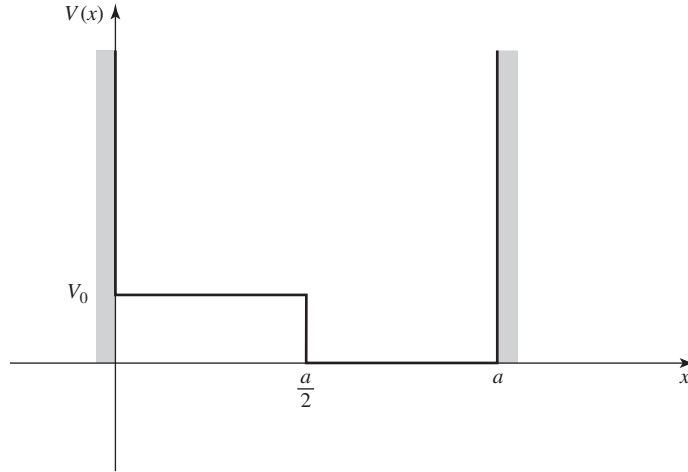


Figure 7.3: Constant perturbation over half the well.

Equation 7.9 is the first-order correction to the *energy*; to find the first-order correction to the *wave function* we rewrite Equation 7.7:

$$(H^0 - E_n^0) \psi_n^1 = -(H' - E_n^1) \psi_n^0. \quad (7.10)$$

The right side is a known function, so this amounts to an inhomogeneous differential equation for ψ_n^1 . Now, the unperturbed wave functions constitute a complete set, so ψ_n^1 (like any other function) can be expressed as a linear combination of them:

$$\psi_n^1 = \sum_{m \neq n} c_m^{(n)} \psi_m^0. \quad (7.11)$$

(There is no need to include $m = n$ in the sum, for if ψ_n^1 satisfies Equation 7.10, so too does $(\psi_n^1 + \alpha \psi_n^0)$, for any constant α , and we can use this freedom to subtract off the ψ_n^0 term.⁴) If we could determine the coefficients $c_m^{(n)}$, we'd be done.

⁴ Alternatively, a glance at Equation 7.5 reveals that any ψ_n^0 component in ψ_n^1 might as well be pulled out and combined with the first term. In fact, the choice $c_n^{(n)} = 0$ ensures that ψ_n —with 1 as the coefficient of ψ_n^0 in Equation 7.5—is *normalized* (to first order in λ): $\langle \psi_n | \psi_n \rangle = \langle \psi_n^0 | \psi_n^0 \rangle + \lambda (\langle \psi_n^1 | \psi_n^0 \rangle + \langle \psi_n^0 | \psi_n^1 \rangle) + \lambda^2 (\dots) + \dots$, but the orthonormality of the unperturbed states means that the first term is 1 and $\langle \psi_n^1 | \psi_n^0 \rangle = \langle \psi_n^0 | \psi_n^1 \rangle = 0$, as long as ψ_n^1 has no ψ_n^0 component.

Well, putting Equation 7.11 into Equation 7.10, and using the fact that ψ_m^0 satisfies the unperturbed Schrödinger equation (Equation 7.1), we have

$$\sum_{m \neq n} (E_m^0 - E_n^0) c_m^{(n)} \psi_m^0 = - (H' - E_n^1) \psi_n^0.$$

Taking the inner product with ψ_l^0 ,

$$\sum_{m \neq n} (E_m^0 - E_n^0) c_m^{(n)} \langle \psi_l^0 | \psi_m^0 \rangle = - \langle \psi_l^0 | H' | \psi_n^0 \rangle + E_n^1 \langle \psi_l^0 | \psi_n^0 \rangle.$$

If $l = n$, the left side is zero, and we recover Equation 7.9; if $l \neq n$, we get

$$(E_l^0 - E_n^0) c_l^{(n)} = - \langle \psi_l^0 | H' | \psi_n^0 \rangle,$$

or

$$c_m^{(n)} = \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle}{E_n^0 - E_m^0}, \quad (7.12)$$

so

$$\psi_n^1 = \sum_{m \neq n} \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle}{(E_n^0 - E_m^0)} \psi_m^0. \quad (7.13)$$

Notice that the denominator is safe (since there is no coefficient with $m = n$) *as long as the unperturbed energy spectrum is nondegenerate*. But if two different unperturbed states share the same energy, we're in serious trouble (we divided by zero to get Equation 7.12); in that case we need **degenerate perturbation theory**, which I'll come to in Section 7.2.

That completes first-order perturbation theory: The first-order correction to the energy, E_n^1 , is given by Equation 7.9, and the first-order correction to the wave function, ψ_n^1 , is given by Equation 7.13.

* **Problem 7.1** Suppose we put a delta-function bump in the center of the infinite square well:

$$H' = \alpha \delta(x - a/2),$$

where α is a constant.

- (a) Find the first-order correction to the allowed energies. Explain why the energies are not perturbed for even n .
- (b) Find the first three nonzero terms in the expansion (Equation 7.13) of the correction to the ground state, ψ_1^1 .

* **Problem 7.2** For the harmonic oscillator [$V(x) = (1/2)kx^2$], the allowed energies are

$$E_n = (n + 1/2)\hbar\omega, \quad (n = 0, 1, 2, \dots),$$

where $\omega = \sqrt{k/m}$ is the classical frequency. Now suppose the spring constant increases slightly: $k \rightarrow (1 + \epsilon)k$. (Perhaps we cool the spring, so it becomes less flexible.)

- (a) Find the *exact* new energies (trivial, in this case). Expand your formula as a power series in ϵ , up to second order.
- (b) Now calculate the first-order perturbation in the energy, using Equation 7.9. What is H' here? Compare your result with part (a). *Hint:* It is not necessary—in fact, it is not *permitted*—to calculate a single integral in doing this problem.

Problem 7.3 Two identical spin-zero bosons are placed in an infinite square well (Equation 2.22). They interact weakly with one another, via the potential

$$V(x_1, x_2) = -a V_0 \delta(x_1 - x_2)$$

(where V_0 is a constant with the dimensions of energy, and a is the width of the well).

- (a) First, ignoring the interaction between the particles, find the ground state and the first excited state—both the wave functions and the associated energies.
- (b) Use first-order perturbation theory to estimate the effect of the particle–particle interaction on the energies of the ground state and the first excited state.

7.1.3 Second-Order Energies

Proceeding as before, we take the inner product of the *second*-order equation (Equation 7.8) with ψ_n^0 :

$$\langle \psi_n^0 | H^0 \psi_n^2 \rangle + \langle \psi_n^0 | H' \psi_n^1 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^2 \rangle + E_n^1 \langle \psi_n^0 | \psi_n^1 \rangle + E_n^2 \langle \psi_n^0 | \psi_n^0 \rangle.$$

Again, we exploit the hermiticity of H^0 :

$$\langle \psi_n^0 | H^0 \psi_n^2 \rangle = \langle H^0 \psi_n^0 | \psi_n^2 \rangle = E_n^0 \langle \psi_n^0 | \psi_n^2 \rangle,$$

so the first term on the left cancels the first term on the right. Meanwhile, $\langle \psi_n^0 | \psi_n^0 \rangle = 1$, and we are left with a formula for E_n^2 :

$$E_n^2 = \langle \psi_n^0 | H' | \psi_n^1 \rangle - E_n^1 \langle \psi_n^0 | \psi_n^1 \rangle. \quad (7.14)$$

But

$$\langle \psi_n^0 | \psi_n^1 \rangle = \sum_{m \neq n} c_m^{(n)} \langle \psi_n^0 | \psi_m^0 \rangle = 0$$

(because the sum excludes $m = n$, and all the others are orthogonal), so

$$E_n^2 = \langle \psi_n^0 | H' | \psi_n^1 \rangle = \sum_{m \neq n} c_m^{(n)} \langle \psi_n^0 | H' | \psi_m^0 \rangle = \sum_{m \neq n} \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle \langle \psi_n^0 | H' | \psi_m^0 \rangle}{E_n^0 - E_m^0},$$

or, finally,

$$E_n^2 = \sum_{m \neq n} \frac{|\langle \psi_m^0 | H' | \psi_n^0 \rangle|^2}{E_n^0 - E_m^0}. \quad (7.15)$$

This is the fundamental result of second-order perturbation theory.

We could go on to calculate the second-order correction to the wave function (ψ_n^2), the third-order correction to the energy, and so on, but in practice Equation 7.15 is ordinarily as far as it is useful to pursue this method.⁵

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Problem 7.4 Apply perturbation theory to the most general two-level system. The unperturbed Hamiltonian is

$$H^0 = \begin{pmatrix} E_a^0 & 0 \\ 0 & E_b^0 \end{pmatrix}$$

and the perturbation is

$$H' = \lambda \begin{pmatrix} V_{aa} & V_{ab} \\ V_{ba} & V_{bb} \end{pmatrix}$$

with $V_{ba} = V_{ab}^*$, V_{aa} and V_{bb} real, so that H is hermitian. As in Section 7.1.1, λ is a constant that will later be set to 1.

- (a) Find the exact energies for this two-level system.
- (b) Expand your result from (a) to second order in λ (and then set λ to 1). Verify that the terms in the series agree with the results from perturbation theory in Sections 7.1.2 and 7.1.3. Assume that $E_b > E_a$.
- (c) Setting $V_{aa} = V_{bb} = 0$, show that the series in (b) only converges if

$$\left| \frac{V_{ab}}{E_b^0 - E_a^0} \right| < \frac{1}{2}.$$

Comment: In general, perturbation theory is only valid if the matrix elements of the perturbation are small compared to the energy level spacings. Otherwise, the first few terms (which are all we ever calculate) will give a poor approximation to the quantity of interest and, as shown here, the series may fail to converge at all, in which case the first few terms tell us nothing.

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Problem 7.5

- (a) Find the second-order correction to the energies (E_n^2) for the potential in Problem 7.1. *Comment:* You can sum the series explicitly, obtaining $-2m(\alpha/\pi\hbar n)^2$ for odd n .

⁵ In the short-hand notation $V_{mn} \equiv \langle \psi_m^0 | H' | \psi_n^0 \rangle$, $\Delta_{mn} \equiv E_m^0 - E_n^0$, the first three corrections to the n th energy are

$$E_n^1 = V_{nn}, \quad E_n^2 = \sum_{m \neq n} \frac{|V_{nm}|^2}{\Delta_{nm}}, \quad E_n^3 = \sum_{l, m \neq n} \frac{V_{nl} V_{lm} V_{mn}}{\Delta_{nl} \Delta_{nm}} - V_{nn} \sum_{m \neq n} \frac{|V_{nm}|^2}{\Delta_{nm}^2}.$$

The third-order correction is given in Landau and Lifschitz, *Quantum Mechanics: Non-Relativistic Theory*, 3rd edn, Pergamon, Oxford (1977), page 136; the fourth and fifth orders (together with a powerful general technique for obtaining the higher orders) are developed by Nicholas Wheeler, *Higher-Order Spectral Perturbation* (unpublished Reed College report, 2000). Illuminating alternative formulations of time-independent perturbation theory include the Dalgarno–Lewis method and the closely related “logarithmic” perturbation theory (see, for example, T. Imbo and U. Sukhatme, *Am. J. Phys.* **52**, 140 (1984), for LPT, and H. Mavromatis, *Am. J. Phys.* **59**, 738 (1991), for Dalgarno–Lewis).

- (b) Calculate the second-order correction to the ground state energy (E_0^2) for the potential in Problem 7.2. Check that your result is consistent with the exact solution.

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Problem 7.6 Consider a charged particle in the one-dimensional harmonic oscillator potential. Suppose we turn on a weak electric field (E), so that the potential energy is shifted by an amount $H' = -qEx$.

- (a) Show that there is no first-order change in the energy levels, and calculate the second-order correction. *Hint:* See Problem 3.39.
- (b) The Schrödinger equation can be solved directly in this case, by a change of variables: $x' \equiv x - (qE/m\omega^2)$. Find the exact energies, and show that they are consistent with the perturbation theory approximation.

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Problem 7.7 Consider a particle in the potential shown in Figure 7.3.

- (a) Find the first-order correction to the ground-state wave function. The first three nonzero terms in the sum will suffice.
- (b) Using the method of Problem 2.61 find (numerically) the ground-state wave function and energy. Use $V_0 = 4\hbar^2/m a^2$ and $N = 100$. Compare the energy obtained numerically to the result from first-order perturbation theory (see Example 7.1).
- (c) Make a single plot showing (i) the unperturbed ground-state wave function, (ii) the numerical ground-state wave function, and (iii) the first-order approximation to the ground-state wave function. *Note:* Make sure you've properly normalized your numerical result,

$$1 = \int |\psi(x)|^2 dx \approx \sum_{i=1}^N |\psi_i|^2 \Delta x.$$

7.2 DEGENERATE PERTURBATION THEORY

If the unperturbed states are degenerate—that is, if two (or more) distinct states (ψ_a^0 and ψ_b^0) share the same energy—then ordinary perturbation theory fails: $c_a^{(b)}$ (Equation 7.12) and E_a^2 (Equation 7.15) blow up (unless, perhaps, the numerator vanishes, $\langle \psi_a^0 | H' | \psi_b^0 \rangle = 0$ —a loophole that will be important to us later on). In the degenerate case, therefore, there is no reason to trust even the *first*-order correction to the energy (Equation 7.9), and we must look for some other way to handle the problem. Note this is *not* a minor problem; almost all applications of perturbation theory involve degeneracy.

7.2.1 Two-Fold Degeneracy

Suppose that

$$H^0 \psi_a^0 = E^0 \psi_a^0, \quad H^0 \psi_b^0 = E^0 \psi_b^0, \quad \langle \psi_a^0 | \psi_b^0 \rangle = 0, \quad (7.16)$$

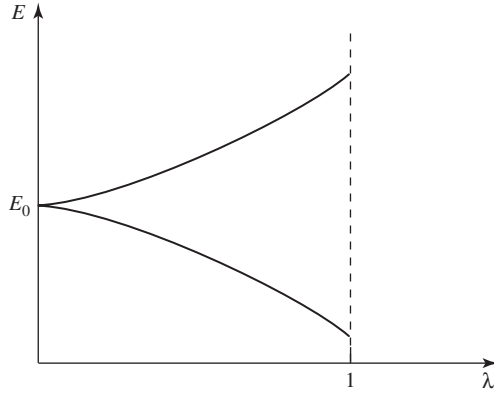


Figure 7.4: “Lifting” of a degeneracy by a perturbation.

with ψ_a^0 and ψ_b^0 both normalized. Note that any linear combination of these states,

$$\psi^0 = \alpha \psi_a^0 + \beta \psi_b^0, \quad (7.17)$$

is still an eigenstate of H^0 , with the same eigenvalue E^0 :

$$H^0 \psi^0 = E^0 \psi^0. \quad (7.18)$$

Typically, the perturbation (H') will “break” (or “lift”) the degeneracy: As we increase λ (from 0 to 1), the common unperturbed energy E^0 splits into two (Figure 7.4). Going the other direction, when we turn *off* the perturbation, the “upper” state reduces down to *one* linear combination of ψ_a^0 and ψ_b^0 , and the “lower” state reduces to some (orthogonal) linear combination, but we don’t know a priori what these “**good**” linear combinations will be. For this reason we can’t even calculate the *first-order* energy (Equation 7.9)—we don’t know what unperturbed states to use.

The “good” states are *defined* as the limit of the true eigenstates as the perturbation is switched off ($\lambda \rightarrow 0$) but that isn’t how you find them in realistic situations (if you *knew* the exact eigenstates you wouldn’t need perturbation theory). Before I show you the practical techniques for calculating them, we’ll look at an example where we can take the $\lambda \rightarrow 0$ limit of the exact eigenstates.

Example 7.2

Consider a particle of mass m in a two-dimensional oscillator potential

$$H^0 = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 (x^2 + y^2)$$

to which is added a perturbation

$$H' = \epsilon m \omega^2 x y.$$

The unperturbed first-excited state (with $E^0 = 2 \hbar \omega$) is two-fold degenerate, and *one* basis for those two degenerate states is

$$\psi_a^0 = \psi_0(x) \psi_1(y) = \sqrt{\frac{2}{\pi}} \frac{m \omega}{\hbar} y e^{-\frac{m \omega}{2 \hbar} (x^2 + y^2)}$$

$$\psi_b^0 = \psi_1(x) \psi_0(y) = \sqrt{\frac{2}{\pi}} \frac{m\omega}{\hbar} x e^{-\frac{m\omega}{2\hbar}(x^2+y^2)}, \quad (7.19)$$

where ψ_0 and ψ_1 refer to the one-dimensional harmonic oscillator states (Equation 2.86). To find the “good” linear combinations, solve for the exact eigenstates of $H = H^0 + H'$ and take their limit as $\epsilon \rightarrow 0$. *Hint:* The problem can be solved by rotating coordinates

$$x' = \frac{x+y}{\sqrt{2}} \quad y' = \frac{x-y}{\sqrt{2}}. \quad (7.20)$$

Solution: In terms of the rotated coordinates, the Hamiltonian is

$$H = \frac{1}{2m} \left(\frac{\partial^2}{\partial x'^2} + \frac{\partial^2}{\partial y'^2} \right) + \frac{1}{2} m (1+\epsilon) \omega^2 x'^2 + \frac{1}{2} m (1-\epsilon) \omega^2 y'^2.$$

This amounts to two independent one-dimensional oscillators. The exact solutions are

$$\psi_{mn} = \psi_m^+(x') \psi_n^-(y'),$$

where $\psi_m^\pm$ are one-dimensional oscillator states with frequencies $\omega_\pm = \sqrt{1 \pm \epsilon} \omega$ respectively. The first few exact energies,

$$E_{mn} = \left(m + \frac{1}{2}\right) \hbar \omega_+ + \left(n + \frac{1}{2}\right) \hbar \omega_-, \quad (7.21)$$

are shown in Figure 7.5.

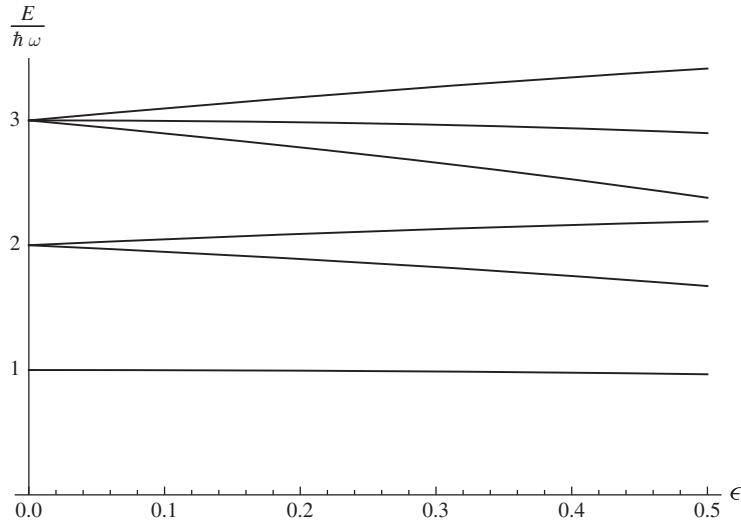


Figure 7.5: Exact energy levels as a function of ϵ for Example 7.2.

The two states which grow out of the degenerate first-excited states as ϵ is increased have $m = 0, n = 1$ (lower state) and $m = 1, n = 0$ (upper state). If we track these states back to $\epsilon = 0$ (in that limit $\omega_+ = \omega_- = \omega$) we get

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \psi_{01}(x) &= \lim_{\epsilon \rightarrow 0} \psi_0^+(x') \psi_1^-(y') = \psi_0\left(\frac{x+y}{\sqrt{2}}\right) \psi_1\left(\frac{x-y}{\sqrt{2}}\right) \\ &= \sqrt{\frac{2}{\pi}} \frac{m\omega}{\hbar} \frac{x-y}{\sqrt{2}} e^{-\frac{m\omega}{2\hbar}(x^2+y^2)} = \frac{-\psi_a^0 + \psi_b^0}{\sqrt{2}}. \end{aligned}$$

$$\lim_{\epsilon \rightarrow 0} \psi_{10}(x) = \frac{\psi_a^0 + \psi_b^0}{\sqrt{2}}. \quad (7.22)$$

Therefore the “good” states for this problem are

$$\psi_{\pm}^0 \equiv \frac{1}{\sqrt{2}} (\psi_b^0 \pm \psi_a^0). \quad (7.23)$$

In this example we were able to find the exact eigenstates of H and then turn off the perturbation to see what states they evolve from. But how do we find the “good” states when we *can't* solve the system exactly?

For the moment let's just write the “good” unperturbed states in generic form (Equation 7.17), keeping α and β adjustable. We want to solve the Schrödinger equation,

$$H\psi = E\psi, \quad (7.24)$$

with $H = H^0 + \lambda H'$ and

$$E = E^0 + \lambda E^1 + \lambda^2 E^2 + \dots, \quad \psi = \psi^0 + \lambda \psi^1 + \lambda^2 \psi^2 + \dots. \quad (7.25)$$

Plugging these into Equation 7.24, and collecting like powers of λ (as before) we find

$$H^0 \psi^0 + \lambda (H' \psi^0 + H^0 \psi^1) + \dots = E^0 \psi^0 + \lambda (E^1 \psi^0 + E^0 \psi^1) + \dots.$$

But $H^0 \psi^0 = E^0 \psi^0$ (Equation 7.18), so the first terms cancel; at order λ^1 we have

$$H^0 \psi^1 + H' \psi^0 = E^0 \psi^1 + E^1 \psi^0. \quad (7.26)$$

Taking the inner product with ψ_a^0 :

$$\langle \psi_a^0 | H^0 \psi^1 \rangle + \langle \psi_a^0 | H' \psi^0 \rangle = E^0 \langle \psi_a^0 | \psi^1 \rangle + E^1 \langle \psi_a^0 | \psi^0 \rangle.$$

Because H^0 is hermitian, the first term on the left cancels the first term on the right. Putting in Equation 7.17, and exploiting the orthonormality condition (Equation 7.16), we obtain

$$\alpha \langle \psi_a^0 | H' | \psi_a^0 \rangle + \beta \langle \psi_a^0 | H' | \psi_b^0 \rangle = \alpha E^1,$$

or, more compactly,

$$\alpha W_{aa} + \beta W_{ab} = \alpha E^1, \quad (7.27)$$

where

$$W_{ij} \equiv \langle \psi_i^0 | H' | \psi_j^0 \rangle. \quad (i, j = a, b). \quad (7.28)$$

Similarly, the inner product with ψ_b^0 yields

$$\alpha W_{ba} + \beta W_{bb} = \beta E^1. \quad (7.29)$$

Notice that the W s are (in principle) *known*—they are just the “matrix elements” of H' , with respect to the unperturbed wave functions ψ_a^0 and ψ_b^0 . Written in matrix form, Equations 7.27 and 7.29 are

$$\underbrace{\begin{pmatrix} W_{aa} & W_{ab} \\ W_{ba} & W_{bb} \end{pmatrix}}_W \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E^1 \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \quad (7.30)$$

The eigenvalues of the matrix W give the first-order corrections to the energy (E^1) and the corresponding eigenvectors tell us the coefficients α and β that determine the “good” states.⁶

The Appendix (Section A.5) shows how to obtain the eigenvalues of a matrix; I’ll reproduce those steps here to find a general solution for E^1 . First, move all the terms in Equation 7.30 to the left-hand side.

$$\begin{pmatrix} W_{aa} - E^1 & W_{ab} \\ W_{ba} & W_{bb} - E^1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0 \quad (7.31)$$

This equation only has non-trivial solutions if the matrix on the left is non-invertible—that is to say, if its determinant vanishes:

$$\begin{vmatrix} W_{aa} - E^1 & W_{ab} \\ W_{ba} & W_{bb} - E^1 \end{vmatrix} = (W_{aa} - E^1)(W_{bb} - E^1) - |W_{ab}|^2 = 0, \quad (7.32)$$

where we used the fact that $W_{ba} = W_{ab}^*$. Solving the quadratic,

$$E_{\pm}^1 = \frac{1}{2} \left[W_{aa} + W_{bb} \pm \sqrt{(W_{aa} - W_{bb})^2 + 4|W_{ab}|^2} \right]. \quad (7.33)$$

This is the fundamental result of degenerate perturbation theory; the two roots correspond to the two perturbed energies.

Example 7.3

Returning to Example 7.2, show that diagonalizing the matrix W gives the same “good” states we found by solving the problem exactly.

Solution: We need to calculate the matrix elements of W . First,

$$\begin{aligned} W_{aa} &= \int \int \psi_a^0(x, y) H' \psi_a^0(x, y) dx dy \\ &= \epsilon m \omega^2 \int |\psi_0(x)|^2 x dx \int |\psi_0(y)|^2 y dy = 0 \end{aligned}$$

(the integrands are both odd functions). Similarly, $W_{bb} = 0$, and we need only compute

$$\begin{aligned} W_{ab} &= \int \int \psi_a^0(x, y) H' \psi_b^0(x, y) dx dy \\ &= \epsilon m \omega^2 \int \psi_0(x) x \psi_1(x) dx \int \psi_1(y) y \psi_0(y) dy. \end{aligned}$$

These two integrals are equal, and recalling (Equation 2.70)

$$x = \sqrt{\frac{\hbar}{2m\omega}} (a_+ + a_-)$$

⁶ This assumes that the eigenvalues of W are *distinct* so that the degeneracy lifts at first order. If not, any choice of α and β satisfies Equation 7.30; you *still* don’t know what the good states are. The first-order energies are correctly given by Equation 7.33 when this happens, and in many cases that’s all you require. But if you need to know the “good” states—for example to calculate higher-order corrections—you will have to use *second-order* degenerate perturbation theory (see Problems 7.39, 7.40, and 7.41) or employ the theorem of Section 7.2.2.

we have

$$\begin{aligned} W_{ab} &= \epsilon m \omega^2 \left[\int \psi_0(x) \sqrt{\frac{\hbar}{2m\omega}} (a_+ + a_-) \psi_1(x) dx \right]^2 \\ &= \epsilon \frac{\hbar \omega}{2} \left[\int \psi_0(x) \psi_0(x) dx \right]^2 = \epsilon \frac{\hbar \omega}{2}. \end{aligned}$$

Therefore, the matrix W is

$$W = \epsilon \frac{\hbar \omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The (normalized) eigenvectors of this matrix are

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

These eigenvectors tell us which linear combination of ψ_a^0 and ψ_b^0 are the good states:

$$\psi_{\pm}^0 = \frac{1}{\sqrt{2}} (\psi_b^0 \pm \psi_a^0),$$

just as in Equation 7.23. The eigenvalues of the matrix W ,

$$E^1 = \pm \epsilon \frac{\hbar \omega}{2}, \quad (7.34)$$

give the first-order corrections to the energy (compare 7.33).

If it happens that $W_{ab} = 0$ in Equation 7.30 then the two eigenvectors are

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and the energies,

$$E_+^1 = W_{aa} = \langle \psi_a^0 | H' | \psi_a^0 \rangle, \quad E_-^1 = W_{bb} = \langle \psi_b^0 | H' | \psi_b^0 \rangle, \quad (7.35)$$

are precisely what we would have obtained using *nondegenerate* perturbation theory (Equation 7.9). We have simply been *lucky*: The states ψ_a^0 and ψ_b^0 were *already* the “good” linear combinations. Obviously, it would be greatly to our advantage if we could somehow *guess* the “good” states right from the start—then we could go ahead and use *nondegenerate* perturbation theory. As it turns out, we can very often do this by exploiting the theorem in the following section.

7.2.2 “Good” States

Theorem: Let A be a hermitian operator that commutes with H^0 and H' . If ψ_a^0 and ψ_b^0 (the degenerate eigenfunctions of H^0) are also eigenfunctions of A , with distinct eigenvalues,

$$A \psi_a^0 = \mu \psi_a^0, \quad A \psi_b^0 = \nu \psi_b^0, \quad \text{and } \mu \neq \nu,$$

then ψ_a^0 and ψ_b^0 are the “good” states to use in perturbation theory.

Proof: Since $H(\lambda) = H^0 + \lambda H'$ and A commute, there exist simultaneous eigenstates $\psi_\gamma(\lambda)$ where

$$H(\lambda) \psi_\gamma(\lambda) = E(\lambda) \psi_\gamma(\lambda) \text{ and } A \psi_\gamma(\lambda) = \gamma \psi_\gamma(\lambda). \quad (7.36)$$

The fact that A is hermitian means

$$\begin{aligned} \langle \psi_a^0 | A \psi_\gamma(\lambda) \rangle &= \langle A \psi_a^0 | \psi_\gamma(\lambda) \rangle \\ \gamma \langle \psi_a^0 | \psi_\gamma(\lambda) \rangle &= \mu^* \langle \psi_a^0 | \psi_\gamma(\lambda) \rangle \end{aligned} \quad (7.37)$$

$$(\gamma - \mu) \langle \psi_a^0 | \psi_\gamma(\lambda) \rangle = 0, \quad (7.38)$$

(making use of the fact that μ is real). This holds true for any value of λ and taking the limit as $\lambda \rightarrow 0$ we have

$$\langle \psi_a^0 | \psi_\gamma(0) \rangle = 0 \text{ unless } \gamma = \mu,$$

and similarly

$$\langle \psi_b^0 | \psi_\gamma(0) \rangle = 0 \text{ unless } \gamma = \nu.$$

Now the good states are linear combinations of ψ_a^0 and ψ_b^0 : $\psi_\gamma(0) = \alpha \psi_a^0 + \beta \psi_b^0$. From above it follows that either $\gamma = \mu$, in which case $\beta = \langle \psi_b^0 | \psi_\gamma(0) \rangle = 0$ and the good state is simply ψ_a^0 , or $\gamma = \nu$ and the good state is ψ_b^0 . QED

Once we identify the “good” states, either by solving Equation 7.30 or by applying this theorem, we can use these “good” states as our unperturbed states and apply ordinary non-degenerate perturbation theory.⁷ In most cases, the operator A will be suggested by symmetry; as you saw in Chapter 6, symmetries are associated with operators that commute with H —precisely what are required to identify the good states.

Example 7.4

Find an operator A that satisfies the requirements of the preceding theorem to construct the “good” states in Examples 7.2 and 7.3.

Solution: The perturbation H' has less symmetry than H^0 . H^0 had continuous rotational symmetry, but $H = H^0 + H'$ is only invariant under rotations by integer multiples of π . For A , take the operator $R(\pi)$ that rotates a function counterclockwise by an angle π . Acting on our states ψ_a and ψ_b we have

$$R(\pi) \psi_a^0(x, y) = \psi_a^0(-x, -y) = -\psi_a^0(x, y),$$

$$R(\pi) \psi_b^0(x, y) = \psi_b^0(-x, -y) = -\psi_b^0(x, y).$$

That’s no good; we need an operator with distinct eigenvalues. How about the operator that interchanges x and y ? This is a reflection about a 45° diagonal of the well. Call this operator D . D commutes with both H^0 and H' , since they are unchanged when you switch x and y . Now,

⁷ Note that the theorem is more general than Equation 7.30. In order to identify the good states from Equation 7.30, the energies $E_\pm^1$ need to be different. In some cases they are the same and the energies of the degenerate states split at second, third, or higher order in perturbation theory. But the *theorem* allows you to identify the good states in every case.

$$\begin{aligned} D \psi_a^0(x, y) &= \psi_a^0(y, x) = \psi_b^0(x, y), \\ D \psi_b^0(x, y) &= \psi_b^0(y, x) = \psi_a^0(x, y). \end{aligned}$$

So our degenerate eigenstates are not eigenstates of D . But we can construct linear combinations that *are*:

$$\psi_{\pm}^0 \equiv \pm \psi_a^0 + \psi_b^0. \quad (7.39)$$

Then

$$D(\pm \psi_a^0 + \psi_b^0) = \pm D \psi_a^0 + D \psi_b^0 = \pm \psi_b^0 + \psi_a^0 = \pm (\pm \psi_a^0 + \psi_b^0).$$

These are “good” states, since they are eigenstates of an operator D with distinct eigenvalues (± 1), and D commutes with both H^0 and H' .

Moral: If you’re faced with degenerate states, look around for some hermitian operator A that commutes with H^0 and H' ; pick as your unperturbed states ones that are simultaneously eigenfunctions of H^0 and A (with *distinct* eigenvalues). Then use *ordinary* first-order perturbation theory. If you can’t find such an operator, you’ll have to resort to Equation 7.33, but in practice this is seldom necessary.

Problem 7.8 Let the two “good” unperturbed states be

$$\psi_{\pm}^0 = \alpha_{\pm} \psi_a^0 + \beta_{\pm} \psi_b^0,$$

where $\alpha_{\pm}$ and $\beta_{\pm}$ are determined (up to normalization) by Equation 7.27 (or Equation 7.29). Show explicitly that

- (a) $\psi_{\pm}^0$ are orthogonal ($\langle \psi_+^0 | \psi_-^0 \rangle = 0$);
- (b) $\langle \psi_+^0 | H' | \psi_-^0 \rangle = 0$;
- (c) $\langle \psi_{\pm}^0 | H' | \psi_{\pm}^0 \rangle = E_{\pm}^1$, with $E_{\pm}^1$ given by Equation 7.33.

Problem 7.9 Consider a particle of mass m that is free to move in a one-dimensional region of length L that closes on itself (for instance, a bead that slides frictionlessly on a circular wire of circumference L , as in Problem 2.46).

(a) Show that the stationary states can be written in the form

$$\psi_n(x) = \frac{1}{\sqrt{L}} e^{2\pi i n x / L}, \quad (-L/2 < x < L/2),$$

where $n = 0, \pm 1, \pm 2, \dots$, and the allowed energies are

$$E_n = \frac{2}{m} \left(\frac{n\pi\hbar}{L} \right)^2.$$

Notice that—with the exception of the ground state ($n = 0$)—these are all doubly degenerate.

(b) Now suppose we introduce the perturbation

$$H' = -V_0 e^{-x^2/a^2},$$

where $a \ll L$. (This puts a little “dimple” in the potential at $x = 0$, as though we bent the wire slightly to make a “trap”.) Find the first-order correction to E_n , using Equation 7.33. *Hint:* To evaluate the integrals, exploit the fact that $a \ll L$ to extend the limits from $\pm L/2$ to $\pm\infty$; after all, H' is essentially zero outside $-a < x < a$.

- (c) What are the “good” linear combinations of ψ_n and ψ_{-n} , for this problem? (*Hint:* use Eq. 7.27.) Show that with these states you get the first-order correction using Equation 7.9.
- (d) Find a hermitian operator A that fits the requirements of the theorem, and show that the simultaneous eigenstates of H^0 and A are precisely the ones you used in (c).

7.2.3 Higher-Order Degeneracy

In the previous section I assumed the degeneracy was two-fold, but it is easy to see how the method generalizes. In the case of n -fold degeneracy, we look for the eigenvalues of the $n \times n$ matrix

$$W_{ij} = \langle \psi_i^0 | H' | \psi_j^0 \rangle. \quad (7.40)$$

For three-fold degeneracy (with degenerate states ψ_a^0 , ψ_b^0 , and ψ_c^0) the first-order corrections to the energies E^1 are the eigenvalues of W , determined by solving

$$\begin{pmatrix} W_{aa} & W_{ab} & W_{ac} \\ W_{ba} & W_{bb} & W_{bc} \\ W_{ca} & W_{cb} & W_{cc} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = E^1 \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}, \quad (7.41)$$

and the “good” states are the corresponding eigenvectors:⁸

$$\psi^0 = \alpha \psi_a^0 + \beta \psi_b^0 + \gamma \psi_c^0. \quad (7.42)$$

Once again, if you can think of an operator A that *commutes* with H^0 and H' , and use the simultaneous eigenfunctions of A and H^0 , then the W matrix will *automatically* be diagonal, and you won’t have to fuss with calculating the off-diagonal elements of W or solving the characteristic equation.⁹ (If you’re nervous about my generalization from two-fold degeneracy to n -fold degeneracy, work Problem 7.13.)

Problem 7.10 Show that the first-order energy corrections computed in Example 7.3 (Equation 7.34) agree with an expansion of the exact solution (Equation 7.21) to first order in ϵ .

Problem 7.11 Suppose we perturb the infinite cubical well (Problem 4.2) by putting a delta function “bump” at the point $(a/4, a/2, 3a/4)$:

$$H' = a^3 V_0 \delta(x - a/4) \delta(y - a/2) \delta(z - 3a/4).$$

Find the first-order corrections to the energy of the ground state and the (triply degenerate) first excited states.

⁸ If the eigenvalues are degenerate, see footnote 6.

⁹ Degenerate perturbation theory amounts to diagonalization of the *degenerate part* of the Hamiltonian; see Problems 7.34 and 7.35.

- * **Problem 7.12** Consider a quantum system with just three linearly independent states. Suppose the Hamiltonian, in matrix form, is

$$H = V_0 \begin{pmatrix} (1 - \epsilon) & 0 & 0 \\ 0 & 1 & \epsilon \\ 0 & \epsilon & 2 \end{pmatrix},$$

where V_0 is a constant, and ϵ is some small number ($\epsilon \ll 1$).

- (a) Write down the eigenvectors and eigenvalues of the *unperturbed* Hamiltonian ($\epsilon = 0$).
- (b) Solve for the *exact* eigenvalues of H . Expand each of them as a power series in ϵ , up to second order.
- (c) Use first- and second-order *non-degenerate* perturbation theory to find the approximate eigenvalue for the state that grows out of the nondegenerate eigenvector of H^0 . Compare the exact result, from (b).
- (d) Use *degenerate* perturbation theory to find the first-order correction to the two initially degenerate eigenvalues. Compare the exact results.

Problem 7.13 In the text I asserted that the first-order corrections to an n -fold degenerate energy are the eigenvalues of the W matrix, and I justified this claim as the “natural” generalization of the case $n = 2$. *Prove* it, by reproducing the steps in Section 7.2.1, starting with

$$\psi^0 = \sum_{j=1}^n \alpha_j \psi_j^0$$

(generalizing Equation 7.17), and ending by showing that the analog to Equation 7.27 can be interpreted as the eigenvalue equation for the matrix W .

7.3 THE FINE STRUCTURE OF HYDROGEN

In our study of the hydrogen atom (Section 4.2) we took the Hamiltonian—called the Bohr Hamiltonian—to be

$$H_{\text{Bohr}} = -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} \quad (7.43)$$

(electron kinetic energy plus Coulombic potential energy). But this is not quite the whole story. We have already learned how to correct for the motion of the nucleus: Just replace m by the reduced mass (Problem 5.1). More significant is the so-called **fine structure**, which is actually due to two distinct mechanisms: a **relativistic correction**, and **spin-orbit coupling**. Compared to the Bohr energies (Equation 4.70), fine structure is a tiny perturbation—smaller by a factor of α^2 , where

$$\alpha \equiv \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036} \quad (7.44)$$

is the famous **fine structure constant**. Smaller still (by another factor of α) is the **Lamb shift**, associated with the quantization of the electric field, and smaller by yet another order of magnitude is the **hyperfine structure**, which is due to the interaction between the magnetic

Table 7.1: *Hierarchy of corrections to the Bohr energies of hydrogen.*

Bohr energies:	of order	$\alpha^2 mc^2$
Fine structure:	of order	$\alpha^4 mc^2$
Lamb shift:	of order	$\alpha^5 mc^2$
Hyperfine splitting:	of order	$(m/m_p)\alpha^4 mc^2$

dipole moments of the electron and the proton. This hierarchy is summarized in Table 7.1. In the present section we will analyze the fine structure of hydrogen, as an application of time-independent perturbation theory.

Problem 7.14

- (a) Express the Bohr energies in terms of the fine structure constant and the rest energy (mc^2) of the electron.
- (b) Calculate the fine structure constant from first principles (i.e., without recourse to the empirical values of ϵ_0 , e , $\hbar$, and c). *Comment:* The fine structure constant is undoubtedly the most fundamental pure (dimensionless) number in all of physics. It relates the basic constants of electromagnetism (the charge of the electron), relativity (the speed of light), and quantum mechanics (Planck's constant). If you can solve part (b), you have the most certain Nobel Prize in history waiting for you. But I wouldn't recommend spending a lot of time on it right now; many smart people have tried, and all (so far) have failed.

7.3.1 The Relativistic Correction

The first term in the Hamiltonian is supposed to represent kinetic energy:

$$T = \frac{1}{2}mv^2 = \frac{p^2}{2m}, \quad (7.45)$$

and the canonical substitution $\mathbf{p} \rightarrow -i\hbar\nabla$ yields the operator

$$T = -\frac{\hbar^2}{2m}\nabla^2. \quad (7.46)$$

But Equation 7.45 is the *classical* expression for kinetic energy; the *relativistic* formula is

$$T = \frac{mc^2}{\sqrt{1 - (v/c)^2}} - mc^2. \quad (7.47)$$

The first term is the *total* relativistic energy (not counting *potential* energy, which we aren't concerned with at the moment), and the second term is the *rest* energy—the *difference* is the energy attributable to motion.

We need to express T in terms of the (relativistic) momentum,

$$p = \frac{mv}{\sqrt{1 - (v/c)^2}}, \quad (7.48)$$

instead of velocity. Notice that

$$p^2 c^2 + m^2 c^4 = \frac{m^2 v^2 c^2 + m^2 c^4 [1 - (v/c)^2]}{1 - (v/c)^2} = \frac{m^2 c^4}{1 - (v/c)^2} = (T + mc^2)^2,$$

so

$$T = \sqrt{p^2 c^2 + m^2 c^4} - mc^2. \quad (7.49)$$

This relativistic equation for kinetic energy reduces (of course) to the classical result (Equation 7.45), in the nonrelativistic limit $p \ll mc$; expanding in powers of the small number (p/mc) , we have

$$\begin{aligned} T &= mc^2 \left[\sqrt{1 + \left(\frac{p}{mc}\right)^2} - 1 \right] = mc^2 \left[1 + \frac{1}{2} \left(\frac{p}{mc}\right)^2 - \frac{1}{8} \left(\frac{p}{mc}\right)^4 \cdots - 1 \right] \\ &= \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} + \cdots. \end{aligned} \quad (7.50)$$

The lowest-order¹⁰ relativistic correction to the Hamiltonian is therefore

$$H'_r = -\frac{p^4}{8m^3 c^2}. \quad (7.51)$$

In first-order perturbation theory, the correction to E_n is given by the expectation value of H'_r in the unperturbed state (Equation 7.9):

$$E_r^1 = \langle H'_r \rangle = -\frac{1}{8m^3 c^2} \langle \psi | p^4 \psi \rangle = -\frac{1}{8m^3 c^2} \langle p^2 \psi | p^2 \psi \rangle. \quad (7.52)$$

Now, the Schrödinger equation (for the unperturbed states) says

$$p^2 \psi = 2m(E - V)\psi, \quad (7.53)$$

and hence¹¹

$$E_r^1 = -\frac{1}{2mc^2} \langle (E - V)^2 \rangle = -\frac{1}{2mc^2} \left[E^2 - 2E \langle V \rangle + \langle V^2 \rangle \right]. \quad (7.54)$$

So far this is entirely general; but we're interested in hydrogen, for which $V(r) = -(1/4\pi\epsilon_0) e^2/r$:

$$E_r^1 = -\frac{1}{2mc^2} \left[E_n^2 + 2E_n \left(\frac{e^2}{4\pi\epsilon_0} \right) \left\langle \frac{1}{r} \right\rangle + \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left\langle \frac{1}{r^2} \right\rangle \right], \quad (7.55)$$

where E_n is the Bohr energy of the state in question.

To complete the job, we need the expectation values of $1/r$ and $1/r^2$, in the (unperturbed) state $\psi_{n\ell m}$ (Equation 4.89). The first is easy (see Problem 7.15):

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{n^2 a}, \quad (7.56)$$

¹⁰ The kinetic energy of the electron in hydrogen is on the order of 10 eV, which is minuscule compared to its rest energy (511,000 eV), so the hydrogen atom is basically nonrelativistic, and we can afford to keep only the lowest-order correction. In Equation 7.50, p is the *relativistic* momentum (Equation 7.48), *not* the classical momentum mv . It is the former that we now associate with the quantum operator $-i\hbar\nabla$, in Equation 7.51.

¹¹ An earlier edition of this book claimed that p^4 is not hermitian for states with $\ell = 0$ (calling into question the maneuver leading to Equation 7.54). That was incorrect— p^4 is hermitian, for all ℓ (see Problem 7.18).

where a is the Bohr radius (Equation 4.72). The second is not so simple to derive (see Problem 7.42), but the answer is¹²

$$\left\langle \frac{1}{r^2} \right\rangle = \frac{1}{(\ell + 1/2) n^3 a^2}. \quad (7.57)$$

It follows that

$$E_r^1 = -\frac{1}{2mc^2} \left[E_n^2 + 2E_n \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{n^2 a} + \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{(\ell + 1/2) n^3 a^2} \right],$$

or, eliminating a (using Equation 4.72) and expressing everything in terms of E_n (using Equation 4.70):

$$E_r^1 = -\frac{(E_n)^2}{2mc^2} \left[\frac{4n}{\ell + 1/2} - 3 \right]. \quad (7.58)$$

Evidently the relativistic correction is smaller than E_n , by a factor of about $E_n/mc^2 = 2 \times 10^{-5}$.

You might have noticed that I used *non*-degenerate perturbation theory in this calculation (Equation 7.52), in spite of the fact that the hydrogen atom is highly degenerate. But the perturbation is spherically symmetric, so it commutes with L^2 and L_z . Moreover, the eigenfunctions of these operators (taken together) have distinct eigenvalues for the n^2 states with a given E_n . Luckily, then, the wave functions $\psi_{n\ell m}$ are the “good” states for this problem (or, as we say, n , ℓ , and m are the **good quantum numbers**), so as it happens the use of nondegenerate perturbation theory was legitimate (see the “Moral” to Section 7.2.1).

From Equation 7.58 we see that *some* of the degeneracy of the n th energy level has lifted. The $(2\ell + 1)$ -fold degeneracy in m remains; as we saw in Example 6.3 it is due to rotational symmetry, a symmetry that remains intact with this perturbation. On the other hand, the “accidental” degeneracy in ℓ has disappeared; since its source is an additional symmetry unique to the $1/r$ potential (see Problem 6.34), we expect that degeneracy to be broken by practically *any* perturbation.

* **Problem 7.15** Use the virial theorem (Problem 4.48) to prove Equation 7.56.

Problem 7.16 In Problem 4.52 you calculated the expectation value of r^s in the state ψ_{321} . Check your answer for the special cases $s = 0$ (trivial), $s = -1$ (Equation 7.56), $s = -2$ (Equation 7.57), and $s = -3$ (Equation 7.66). Comment on the case $s = -7$.

** **Problem 7.17** Find the (lowest-order) relativistic correction to the energy levels of the one-dimensional harmonic oscillator. *Hint:* Use the technique of Problem 2.12.

¹² The general formula for the expectation value of *any* power of r is given in Hans A. Bethe and Edwin E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms*, Plenum, New York (1977), p. 17.

Problem 7.18 Show that p^2 is hermitian, for hydrogen states with $\ell = 0$. *Hint:* For such states ψ is independent of θ and ϕ , so

$$p^2 = -\frac{\hbar^2}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right)$$

(Equation 4.13). Using integration by parts, show that

$$\langle f | p^2 g \rangle = -4\pi\hbar^2 \left(r^2 f \frac{dg}{dr} - r^2 g \frac{df}{dr} \right) \Big|_0^\infty + \langle p^2 f | g \rangle.$$

Check that the boundary term vanishes for ψ_{n00} , which goes like

$$\psi_{n00} \sim \frac{1}{\sqrt{\pi} (na)^{3/2}} \exp(-r/na)$$

near the origin.

The case of p^4 is more subtle. The Laplacian of $1/r$ picks up a delta function (see, for example, D. J. Griffiths, *Introduction to Electrodynamics*, 4th edn, Eq. 1.102). Show that

$$\nabla^4 [e^{-kr}] = \left(-\frac{4k^3}{r} + k^4 \right) e^{-kr} + 8\pi k \delta^3(\mathbf{r}),$$

and confirm that p^4 is hermitian.¹³

7.3.2 Spin-Orbit Coupling

Imagine the electron in orbit around the nucleus; from the *electron's* point of view, the proton is circling around *it* (Figure 7.6). This orbiting positive charge sets up a magnetic field $\mathbf{B}$, in the electron frame, which exerts a torque on the spinning electron, tending to align its magnetic moment ($\boldsymbol{\mu}$) along the direction of the field. The Hamiltonian (Equation 4.157) is

$$H = -\boldsymbol{\mu} \cdot \mathbf{B}. \quad (7.59)$$

To begin with, we need to figure out the magnetic field of the proton ($\mathbf{B}$) and the dipole moment of the electron ($\boldsymbol{\mu}$).

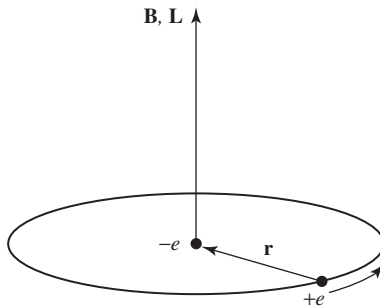


Figure 7.6: Hydrogen atom, from the electron's perspective.

¹³ Thanks to Edward Ross and Li Yi-ding for fixing this problem.

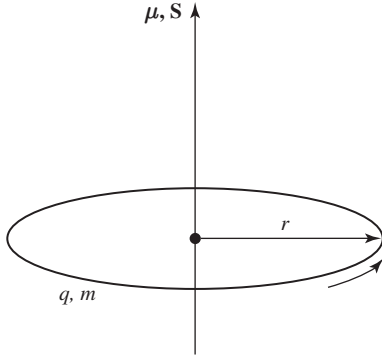


Figure 7.7: A ring of charge, rotating about its axis.

The Magnetic Field of the Proton. If we picture the proton (from the electron's perspective) as a continuous current loop (Figure 7.6), its magnetic field can be calculated from the Biot–Savart law:

$$B = \frac{\mu_0 I}{2r},$$

with an effective current $I = e/T$, where e is the charge of the proton and T is the period of the orbit. On the other hand, the orbital angular momentum of the *electron* (in the rest frame of the *nucleus*) is $L = rmv = 2\pi mr^2/T$. Moreover, $\mathbf{B}$ and $\mathbf{L}$ point in the same direction (up, in Figure 7.6), so

$$\mathbf{B} = \frac{1}{4\pi\epsilon_0} \frac{e}{mc^2 r^3} \mathbf{L}. \quad (7.60)$$

(I used $c = 1/\sqrt{\epsilon_0\mu_0}$ to eliminate μ_0 in favor of ϵ_0 .)

The Magnetic Dipole Moment of the Electron. The magnetic dipole moment of a spinning charge is related to its (spin) angular momentum; the proportionality factor is the gyromagnetic ratio (which we already encountered in Section 4.4.2). Let's derive it, this time, using classical electrodynamics. Consider first a charge q smeared out around a ring of radius r , which rotates about the axis with period T (Figure 7.7). The magnetic dipole moment of the ring is defined as the current (q/T) times the area (πr^2):

$$\mu = \frac{q\pi r^2}{T}.$$

If the mass of the ring is m , its angular momentum is the moment of inertia (mr^2) times the angular velocity ($2\pi/T$):

$$S = \frac{2\pi mr^2}{T}.$$

The gyromagnetic ratio for this configuration is evidently $\mu/S = q/2m$. Notice that it is independent of r (and T). If I had some more complicated object, such as a sphere (all I require is that it be a figure of revolution, rotating about its axis), I could calculate μ and S by chopping it into little rings, and adding up their contributions. As long as the mass and the charge are distributed in the same manner (so that the charge-to-mass ratio is uniform), the gyromagnetic ratio will be the same for each ring, and hence also for the object as a whole. Moreover, the directions of μ and S are the same (or opposite, if the charge is negative), so

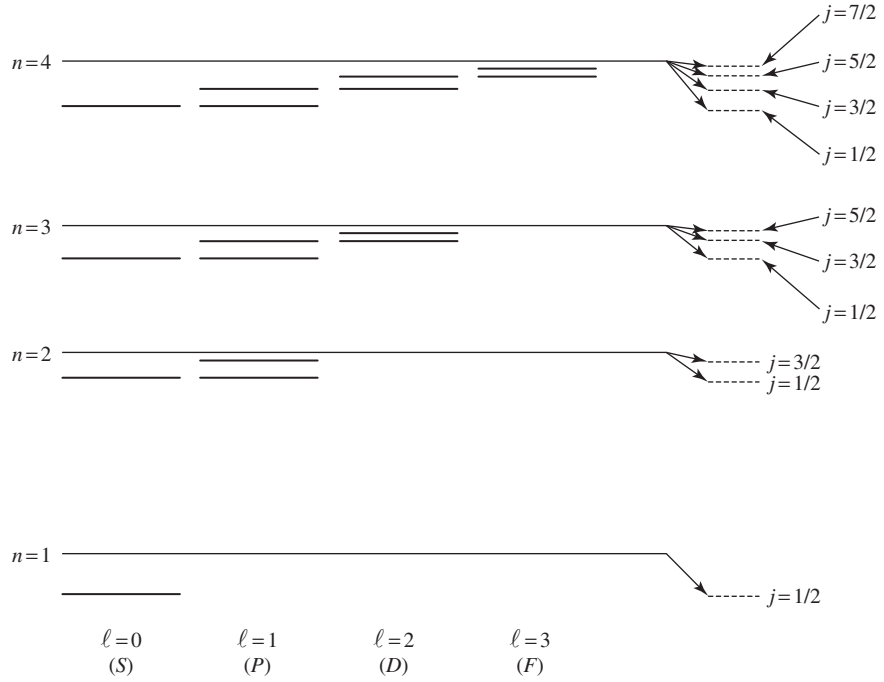


Figure 7.8: Energy levels of hydrogen, including fine structure (not to scale).

$$\boldsymbol{\mu} = \left(\frac{q}{2m} \right) \mathbf{S}. \quad (7.61)$$

That was a purely *classical* calculation, however; as it turns out the electron’s magnetic moment is *twice* the classical value:

$$\boldsymbol{\mu}_e = -\frac{e}{m} \mathbf{S}. \quad (7.62)$$

The “extra” factor of 2 was explained by Dirac, in his relativistic theory of the electron.¹⁴

Putting all this together, we have

$$H = \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{m^2 c^2 r^3} \mathbf{S} \cdot \mathbf{L}.$$

But there is a serious fraud in this calculation: I did the analysis in the rest frame of the electron, but that’s *not an inertial system*—it *accelerates*, as the electron orbits around the nucleus. You

¹⁴ We have already noted that it can be dangerous to picture the electron as a spinning sphere (see Problem 4.28), and it is not too surprising that the naive classical model gets the gyromagnetic ratio wrong. The *deviation* from the classical expectation is known as the **g-factor**: $\boldsymbol{\mu} = g (q/2m) \mathbf{S}$. Thus the *g*-factor of the electron, in Dirac’s theory, is exactly 2. But **quantum electrodynamics** reveals tiny corrections to this: g_e is actually $2 + (\alpha/\pi) + \dots = 2.002\dots$. The calculation and measurement (which agree to exquisite precision) of the so-called **anomalous magnetic moment** of the electron were among the greatest achievements of twentieth-century physics.

can get away with this if you make an appropriate kinematic correction, known as the **Thomas precession**.¹⁵ In this context it throws in a factor of $1/2$.¹⁶

$$H'_{\text{so}} = \left(\frac{e^2}{8\pi\epsilon_0} \right) \frac{1}{m^2 c^2 r^3} \mathbf{S} \cdot \mathbf{L}. \quad (7.63)$$

This is the **spin-orbit interaction**; apart from two corrections (the modified gyromagnetic ratio for the electron and the Thomas precession factor—which, coincidentally, exactly cancel one another) it is just what you would expect on the basis of a naive classical model. Physically, it is due to the torque exerted on the magnetic dipole moment of the spinning electron, by the magnetic field of the proton, in the electron's instantaneous rest frame.

Now the quantum mechanics. In the presence of spin-orbit coupling, the Hamiltonian no longer commutes with $\mathbf{L}$ and $\mathbf{S}$, so the spin and orbital angular momenta are not separately conserved (see Problem 7.19). However, H'_{so} *does* commute with L^2 , S^2 and the *total* angular momentum

$$\mathbf{J} \equiv \mathbf{L} + \mathbf{S}, \quad (7.64)$$

and hence these quantities *are* conserved (Equation 3.73). To put it another way, the eigenstates of L_z and S_z are not “good” states to use in perturbation theory, but the eigenstates of L^2 , S^2 , J^2 , and J_z *are*. Now

$$J^2 = (\mathbf{L} + \mathbf{S}) \cdot (\mathbf{L} + \mathbf{S}) = L^2 + S^2 + 2\mathbf{L} \cdot \mathbf{S},$$

so

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2} (J^2 - L^2 - S^2), \quad (7.65)$$

and therefore the eigenvalues of $\mathbf{L} \cdot \mathbf{S}$ are

$$\frac{\hbar^2}{2} [j(j+1) - \ell(\ell+1) - s(s+1)].$$

In this case, of course, $s = 1/2$. Meanwhile, the expectation value of $1/r^3$ (see Problem 7.43)¹⁷ is

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{1}{\ell(\ell+1/2)(\ell+1)n^3 a^3}, \quad (7.66)$$

¹⁵ One way of thinking of it is that the electron is continually stepping from one inertial system to another; Thomas precession amounts to the cumulative effect of all these Lorentz transformations. We could avoid the whole problem, of course, by staying in the *lab* frame, in which the nucleus is at rest. In that case the field of the proton is purely *electric*, and you might wonder why it exerts any torque on the electron. Well, the fact is that a moving *magnetic* dipole acquires an *electric* dipole moment, and in the lab frame the spin-orbit coupling is due to the interaction of the *electric* field of the nucleus with the *electric* dipole moment of the electron. Because this analysis requires more sophisticated electrodynamics, it seems best to adopt the electron's perspective, where the physical mechanism is more transparent.

¹⁶ More precisely, Thomas precession subtracts 1 from the gyromagnetic ratio (see R. R. Haer and L. J. Curtis, *Am. J. Phys.*, **55**, 1044 (1987)).

¹⁷ In Problem 7.43 the expectation values are calculated using the hydrogen wave functions $\psi(n\ell m)$ —that is, eigenstates of L_z —whereas we now want eigenstates of J_z —which are linear combinations of $m = m_j + \frac{1}{2}$ and $m = m_j - \frac{1}{2}$. But since $\langle r^s \rangle$ is independent of m , it doesn't matter.

and we conclude that

$$E_{\text{so}}^1 = \langle H'_{\text{so}} \rangle = \frac{e^2}{8\pi\epsilon_0} \frac{1}{m^2 c^2} \frac{(\hbar^2/2) [j(j+1) - \ell(\ell+1) - 3/4]}{\ell(\ell+1/2)(\ell+1)n^3 a^3},$$

or, expressing it all in terms of E_n :¹⁸

$$E_{\text{so}}^1 = \frac{(E_n)^2}{mc^2} \left\{ \frac{n [j(j+1) - \ell(\ell+1) - 3/4]}{\ell(\ell+1/2)(\ell+1)} \right\}. \quad (7.67)$$

It is remarkable, considering the totally different physical mechanisms involved, that the relativistic correction and the spin-orbit coupling are of the same order (E_n^2/mc^2). Adding them together, we get the complete fine-structure formula (see Problem 7.20):

$$E_{\text{fs}}^1 = \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j+1/2} \right). \quad (7.68)$$

Combining this with the Bohr formula, we obtain the grand result for the energy levels of hydrogen, including fine structure:

$$E_{nj} = -\frac{13.6 \text{ eV}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j+1/2} - \frac{3}{4} \right) \right]. \quad (7.69)$$

Fine structure breaks the degeneracy in ℓ (that is, for a given n , the different allowed values of ℓ do not all carry the same energy), but it still preserves degeneracy in j (see Figure 7.8). The z -component eigenvalues for orbital and spin angular momentum (m_ℓ and m_s) are no longer “good” quantum numbers—the stationary states are linear combinations of states with different values of these quantities; the “good” quantum numbers are n , ℓ , s , j , and m_j .¹⁹

Problem 7.19 Evaluate the following commutators: (a) $[\mathbf{L} \cdot \mathbf{S}, \mathbf{L}]$, (b) $[\mathbf{L} \cdot \mathbf{S}, \mathbf{S}]$, (c) $[\mathbf{L} \cdot \mathbf{S}, \mathbf{J}]$, (d) $[\mathbf{L} \cdot \mathbf{S}, L^2]$, (e) $[\mathbf{L} \cdot \mathbf{S}, S^2]$, (f) $[\mathbf{L} \cdot \mathbf{S}, J^2]$. *Hint:* $\mathbf{L}$ and $\mathbf{S}$ satisfy the fundamental commutation relations for angular momentum (Equations 4.99 and 4.134), but they commute with each other.

¹⁸ The case $\ell = 0$ looks problematic, since we are ostensibly dividing by zero. On the other hand, the numerator is *also* zero, since in this case $j = s$; so Equation 7.67 is indeterminate. On physical grounds there shouldn't be any spin-orbit coupling when $\ell = 0$. In any event, the problem disappears when the spin-orbit coupling is added to the relativistic correction, and their *sum* (Equation 7.68) is correct for *all* ℓ . If you're feeling uneasy about this whole calculation, I don't blame you; take comfort in the fact that the *exact* solution can be obtained by using the (relativistic) Dirac equation in place of the (nonrelativistic) Schrödinger equation, and it confirms the results we obtain here by less rigorous means (see Problem 7.22.)

¹⁹ To write $|jm_j\rangle$ (for given ℓ and s) as a linear combination of $|\ell s m_\ell m_s\rangle$ we would use the appropriate Clebsch–Gordan coefficients (Equation 4.183).

- * **Problem 7.20** Derive the fine structure formula (Equation 7.68) from the relativistic correction (Equation 7.58) and the spin-orbit coupling (Equation 7.67). *Hint:* Note that $j = \ell \pm 1/2$ (except for $\ell = 0$, where only the plus sign occurs); treat the plus sign and the minus sign separately, and you'll find that you get the same final answer either way.

- ** **Problem 7.21** The most prominent feature of the hydrogen spectrum in the visible region is the red Balmer line, coming from the transition $n = 3$ to $n = 2$. First of all, determine the wavelength and frequency of this line according to the Bohr theory. Fine structure splits this line into several closely-spaced lines; the question is: *How many*, and *what is their spacing?* *Hint:* First determine how many sublevels the $n = 2$ level splits into, and find E_{fs}^1 for each of these, in eV. Then do the same for $n = 3$. Draw an energy level diagram showing all possible transitions from $n = 3$ to $n = 2$. The energy released (in the form of a photon) is $(E_3 - E_2) + \Delta E$, the first part being common to all of them, and ΔE (due to fine structure) varying from one transition to the next. Find ΔE (in eV) for each transition. Finally, convert to photon frequency, and determine the spacing between adjacent spectral lines (in Hz)—*not* the frequency interval between each line and the *unperturbed* line (which is, of course, unobservable), but the frequency interval between each line and the *next* one. Your final answer should take the form: “The red Balmer line splits into (???) lines. In order of increasing frequency, they come from the transitions (1) $j = (???)$ to $j = (???)$, (2) $j = (???)$ to $j = (???)$, The frequency spacing between line (1) and line (2) is (???) Hz, the spacing between line (2) and line (3) is (???) Hz, ...”

Problem 7.22 The *exact* fine-structure formula for hydrogen (obtained from the Dirac equation without recourse to perturbation theory) is²⁰

$$E_{nj} = mc^2 \left\{ \left[1 + \left(\frac{\alpha}{n - (j + 1/2) + \sqrt{(j + 1/2)^2 - \alpha^2}} \right)^2 \right]^{-1/2} - 1 \right\}.$$

Expand to order α^4 (noting that $\alpha \ll 1$), and show that you recover Equation 7.69.

7.4 THE ZEEMAN EFFECT

When an atom is placed in a uniform external magnetic field $\mathbf{B}_{\text{ext}}$, the energy levels are shifted. This phenomenon is known as the **Zeeman effect**. For a single electron, the perturbation is²¹

$$H'_Z = -(\boldsymbol{\mu}_l + \boldsymbol{\mu}_s) \cdot \mathbf{B}_{\text{ext}}, \quad (7.70)$$

²⁰ Bethe and Salpeter (footnote 12, page 298), page 238.

²¹ This is correct to first order in B . We are ignoring a term of order B^2 in the Hamiltonian (the exact result was calculated in Problem 4.72). In addition, the orbital magnetic moment (Equation 7.72) is proportional to the mechanical angular momentum, not the canonical angular momentum (see Problem 7.49). These neglected terms give corrections of order B^2 , comparable to the second-order corrections from H'_Z . Since we're working to first order, they are safe to ignore in this context.

where

$$\mu_s = -\frac{e}{m}\mathbf{S} \quad (7.71)$$

(Equation 7.62) is the magnetic dipole moment associated with electron spin, and

$$\mu_l = -\frac{e}{2m}\mathbf{L} \quad (7.72)$$

(Equation 7.61) is the dipole moment associated with orbital motion.²² Thus

$$H'_Z = \frac{e}{2m}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B}_{\text{ext}}. \quad (7.73)$$

The nature of the Zeeman splitting depends critically on the strength of the external field in comparison with the *internal* field (Equation 7.60) that gives rise to spin-orbit coupling. If $B_{\text{ext}} \ll B_{\text{int}}$, then fine structure dominates, and H'_Z can be treated as a small perturbation, whereas if $B_{\text{ext}} \gg B_{\text{int}}$, then the Zeeman effect dominates, and fine structure becomes the perturbation. In the intermediate zone, where the two fields are comparable, we need the full machinery of degenerate perturbation theory, and it is necessary to diagonalize the relevant portion of the Hamiltonian “by hand.” In the following sections we shall explore each of these regimes briefly, for the case of hydrogen.

Problem 7.23 Use Equation 7.60 to estimate the internal field in hydrogen, and characterize quantitatively a “strong” and “weak” Zeeman field.

7.4.1 Weak-Field Zeeman Effect

If $B_{\text{ext}} \ll B_{\text{int}}$, fine structure dominates; we treat $H_{\text{Bohr}} + H'_{fs}$ as the “unperturbed” Hamiltonian and H'_Z as the perturbation. Our “unperturbed” eigenstates are then those appropriate to fine structure: $|n\ell jm_j\rangle$ and the “unperturbed” energies are E_{nj} (Equation 7.69). Even though fine structure has lifted some of the degeneracy in the Bohr model, these states are still degenerate, since the energy does not depend on m_j or ℓ . Luckily the states $|n\ell jm_j\rangle$ are the “good” states for treating the perturbation H'_Z (meaning we don’t have to write down the W matrix for H'_Z —it’s already diagonal) since H'_Z commutes with J_z (so long as we align $\mathbf{B}_{\text{ext}}$ with the z axis) and L^2 , and each of the degenerate states is *uniquely* labeled by the two quantum numbers m_j and ℓ .

In first-order perturbation theory, the Zeeman correction to the energy is

$$E_Z^1 = \langle n\ell jm_j | H'_Z | n\ell jm_j \rangle = \frac{e}{2m} B_{\text{ext}} \hat{k} \cdot \langle \mathbf{L} + 2\mathbf{S} \rangle, \quad (7.74)$$

where, as mentioned above, we align $\mathbf{B}_{\text{ext}}$ with the z axis to eliminate the off-diagonal elements of W . Now $\mathbf{L} + 2\mathbf{S} = \mathbf{J} + \mathbf{S}$. Unfortunately, we do not immediately know the expectation value of $\mathbf{S}$. But we can figure it out, as follows: The total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is constant (Figure 7.9); $\mathbf{L}$ and $\mathbf{S}$ precess rapidly about this fixed vector. In particular, the (time) *average* value of $\mathbf{S}$ is just its projection along $\mathbf{J}$:

$$\mathbf{S}_{\text{ave}} = \frac{(\mathbf{S} \cdot \mathbf{J})}{J^2} \mathbf{J}. \quad (7.75)$$

²² The gyromagnetic ratio for *orbital* motion is just the classical value ($q/2m$)—it is only for *spin* that there is an “extra” factor of 2.

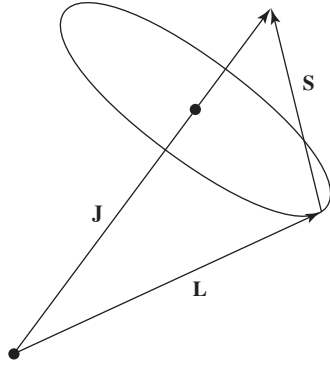


Figure 7.9: In the presence of spin-orbit coupling, $\mathbf{L}$ and $\mathbf{S}$ are not separately conserved; they precess about the fixed total angular momentum, $\mathbf{J}$.

But $\mathbf{L} = \mathbf{J} - \mathbf{S}$, so $L^2 = J^2 + S^2 - 2\mathbf{J} \cdot \mathbf{S}$, and hence

$$\mathbf{S} \cdot \mathbf{J} = \frac{1}{2} (J^2 + S^2 - L^2) = \frac{\hbar^2}{2} [j(j+1) + s(s+1) - \ell(\ell+1)], \quad (7.76)$$

from which it follows that²³

$$\langle \mathbf{L} + 2\mathbf{S} \rangle = \left\langle \left(1 + \frac{\mathbf{S} \cdot \mathbf{J}}{J^2} \right) \mathbf{J} \right\rangle = \left[1 + \frac{j(j+1) - \ell(\ell+1) + s(s+1)}{2j(j+1)} \right] \langle \mathbf{J} \rangle. \quad (7.78)$$

The term in square brackets is known as the **Landé g-factor**, g_J .²⁴

The energy corrections are then

$$E_Z^1 = \mu_B g_J B_{\text{ext}} m_j, \quad (7.79)$$

where

$$\mu_B \equiv \frac{e\hbar}{2m} = 5.788 \times 10^{-5} \text{ eV/T} \quad (7.80)$$

is the so-called **Bohr magneton**. Recall (Example 6.3) that degeneracy in the quantum number m is a consequence of rotational invariance.²⁵ The perturbation H'_Z picks out a specific direction in space (the direction of $\mathbf{B}$) which breaks the rotational symmetry and lifts the degeneracy in m .

²³ While Equation 7.78 was derived by replacing $\mathbf{S}$ by its average value, the result is not an approximation; $\mathbf{L} + 2\mathbf{S}$ and $\mathbf{J}$ are both vector operators and the states are angular-momentum eigenstates. Therefore, the matrix elements can be evaluated by use of the Wigner–Eckart theorem (Equations 6.59–6.61). It follows (Problem 7.25) that the matrix elements are proportional:

$$\langle n\ell j m_j | \mathbf{L} + 2\mathbf{S} | n\ell j m'_j \rangle = g_J \langle n\ell j m_j | \mathbf{J} | n\ell j m'_j \rangle \quad (7.77)$$

and the constant of proportionality g_J is the ratio of reduced matrix elements. All that remains is to evaluate g_J : see Claude Cohen-Tannoudji, Bernard Diu, and Franck Lalœ, *Quantum Mechanics*, Wiley, New York (1977), Vol. 2, Chapter X.

²⁴ In the case of a single electron, where $j = \ell \pm \frac{1}{2}$, $g_J = (2j + 1) / (2\ell + 1)$.

²⁵ That example specifically treated orbital angular momentum, but the same argument holds for the total angular momentum.

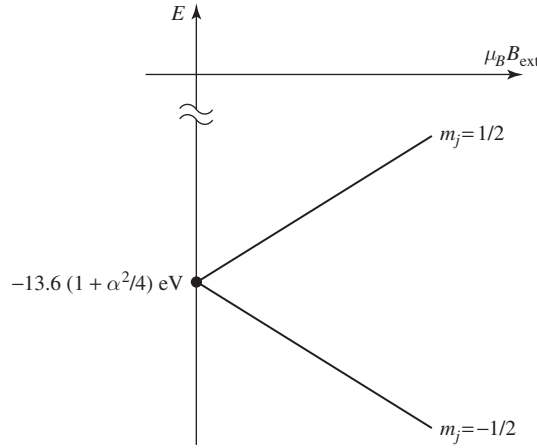


Figure 7.10: Weak-field Zeeman splitting of the ground state of hydrogen; the upper line ($m_j = 1/2$) has slope 1, the lower line ($m_j = -1/2$) has slope -1 .

The *total* energy is the sum of the fine-structure part (Equation 7.69) and the Zeeman contribution (Equation 7.79). For example, the ground state ($n = 1$, $\ell = 0$, $j = 1/2$, and therefore $g_J = 2$) splits into two levels:

$$-13.6 \text{ eV} \left(1 + \alpha^2/4\right) \pm \mu_B B_{\text{ext}}, \quad (7.81)$$

with the plus sign for $m_j = 1/2$, and minus for $m_j = -1/2$. These energies are plotted (as functions of B_{ext}) in Figure 7.10.

*

Problem 7.24 Consider the (eight) $n = 2$ states, $|2\ell j m_j\rangle$. Find the energy of each state, under weak-field Zeeman splitting, and construct a diagram like Figure 7.10 to show how the energies evolve as B_{ext} increases. Label each line clearly, and indicate its slope.

Problem 7.25 Use the Wigner–Eckart theorem (Equations 6.59–6.61) to prove that the matrix elements of any two vector operators, $\mathbf{V}$ and $\mathbf{W}$, are proportional in a basis of angular-momentum eigenstates:

$$\langle n \ell' m' | \mathbf{V} | n \ell m \rangle = \alpha \langle n \ell' m' | \mathbf{W} | n \ell m \rangle. \quad (7.82)$$

Comment: With ℓ replaced by j (the theorem holds regardless of whether the states are eigenstates of orbital, spin, or total angular momentum), $\mathbf{V} = \mathbf{L} + 2\mathbf{S}$ and $\mathbf{W} = \mathbf{J}$, this proves Equation 7.77.

7.4.2 Strong-Field Zeeman Effect

If $B_{\text{ext}} \gg B_{\text{int}}$, the Zeeman effect dominates²⁶ and we take the “unperturbed” Hamiltonian to be $H_{\text{Bohr}} + H'_Z$ and the perturbation to be H'_{fs} . The Zeeman Hamiltonian is

²⁶ In this regime the Zeeman effect is also known as the **Paschen–Back effect**.

$$H'_Z = \frac{e}{2m} B_{\text{ext}} (L_z + 2S_z),$$

and it is straightforward to compute the “unperturbed” energies:

$$E_{n\ell m_\ell m_s} = -\frac{13.6 \text{ eV}}{n^2} + \mu_B B_{\text{ext}} (m_\ell + 2m_s). \quad (7.83)$$

The states we are using here: $|n\ell m_\ell m_s\rangle$ are degenerate, since the energy does not depend on ℓ , and there is an additional degeneracy due to the fact that, for example, $m_\ell = 3$ and $m_s = -1/2$ or $m_\ell = 1$ and $m_s = 1/2$ have the same energy. Again we are lucky; $|n\ell m_\ell m_s\rangle$ are the “good” states for treating the perturbation. The fine structure Hamiltonian H'_{fs} commutes with both L^2 and with J_z (these two operators serve as A in the theorem of Section 7.2.2); the first operator resolves the degeneracy in ℓ and the second resolves the degeneracy from coincidences in $m_\ell + 2m_s = m_j + m_s$.

In first-order perturbation theory the fine structure correction to these levels is

$$E_{\text{fs}}^1 = \langle n\ell m_\ell m_s | (H'_r + H'_{\text{so}}) | n\ell m_\ell m_s \rangle. \quad (7.84)$$

The relativistic contribution is the same as before (Equation 7.58); for the spin-orbit term (Equation 7.63) we need

$$\langle \mathbf{S} \cdot \mathbf{L} \rangle = \langle S_x \rangle \langle L_x \rangle + \langle S_y \rangle \langle L_y \rangle + \langle S_z \rangle \langle L_z \rangle = \hbar^2 m_\ell m_s \quad (7.85)$$

(note that $\langle S_x \rangle = \langle S_y \rangle = \langle L_x \rangle = \langle L_y \rangle = 0$ for eigenstates of S_z and L_z). Putting all this together (Problem 7.26), we conclude that

$$E_{\text{fs}}^1 = \frac{13.6 \text{ eV}}{n^3} \alpha^2 \left\{ \frac{3}{4n} - \left[\frac{\ell(\ell+1) - m_\ell m_s}{\ell(\ell+1/2)(\ell+1)} \right] \right\}. \quad (7.86)$$

(The term in square brackets is indeterminate for $\ell = 0$; its correct value in this case is 1—see Problem 7.28.) The *total* energy is the sum of the Zeeman part (Equation 7.83) and the fine structure contribution (Equation 7.86).

Problem 7.26 Starting with Equation 7.84, and using Equations 7.58, 7.63, 7.66, and 7.85, derive Equation 7.86.

**

Problem 7.27 Consider the (eight) $n = 2$ states, $|2\ell m_\ell m_s\rangle$. Find the energy of each state, under strong-field Zeeman splitting. Express each answer as the sum of three terms: the Bohr energy, the fine-structure (proportional to α^2), and the Zeeman contribution (proportional to $\mu_B B_{\text{ext}}$). If you ignore fine structure altogether, how many distinct levels are there, and what are their degeneracies?

Problem 7.28 If $\ell = 0$, then $j = s$, $m_j = m_s$, and the “good” states are the same ($|n m_s\rangle$) for weak *and* strong fields. Determine E_Z^1 (from Equation 7.74) and the fine structure energies (Equation 7.69), and write down the general result for the $\ell = 0$ Zeeman effect—*regardless* of the strength of the field. Show that the strong-field formula (Equation 7.86) reproduces this result, provided that we interpret the indeterminate term in square brackets as 1.

7.4.3 Intermediate-Field Zeeman Effect

In the intermediate regime neither H'_Z nor H'_{fs} dominates, and we must treat the two on an equal footing, as perturbations to the Bohr Hamiltonian (Equation 7.43):

$$H' = H'_Z + H'_{fs}. \quad (7.87)$$

I'll confine my attention here to the case $n = 2$ (you get to do $n = 3$ in Problem 7.30). It's not obvious what the “good” states are, so we'll have to resort to the full machinery of degenerate perturbation theory. I'll choose basis states characterized by ℓ , j , and m_j .²⁷ Using the Clebsch–Gordan coefficients (Problem 4.60 or Table 4.8) to express $|j m_j\rangle$ as linear combinations of $|\ell s m_\ell m_s\rangle$,²⁸ we have:

$\ell = 0$:

$$\psi_1 \equiv \left| \frac{1}{2} \frac{1}{2} \right\rangle = \left| 0 \frac{1}{2} 0 \frac{1}{2} \right\rangle,$$

$$\psi_2 \equiv \left| \frac{1}{2} \frac{-1}{2} \right\rangle = \left| 0 \frac{1}{2} 0 \frac{-1}{2} \right\rangle,$$

$\ell = 1$:

$$\psi_3 \equiv \left| \frac{3}{2} \frac{3}{2} \right\rangle = \left| 1 \frac{1}{2} 1 \frac{1}{2} \right\rangle,$$

$$\psi_4 \equiv \left| \frac{3}{2} \frac{-3}{2} \right\rangle = \left| 1 \frac{1}{2} -1 \frac{-1}{2} \right\rangle,$$

$$\psi_5 \equiv \left| \frac{3}{2} \frac{1}{2} \right\rangle = \sqrt{2/3} \left| 1 \frac{1}{2} 0 \frac{1}{2} \right\rangle + \sqrt{1/3} \left| 1 \frac{1}{2} 1 \frac{-1}{2} \right\rangle,$$

$$\psi_6 \equiv \left| \frac{1}{2} \frac{1}{2} \right\rangle = -\sqrt{1/3} \left| 1 \frac{1}{2} 0 \frac{1}{2} \right\rangle + \sqrt{2/3} \left| 1 \frac{1}{2} 1 \frac{-1}{2} \right\rangle,$$

$$\psi_7 \equiv \left| \frac{3}{2} \frac{-1}{2} \right\rangle = \sqrt{1/3} \left| 1 \frac{1}{2} -1 \frac{1}{2} \right\rangle + \sqrt{2/3} \left| 1 \frac{1}{2} 0 \frac{-1}{2} \right\rangle,$$

$$\psi_8 \equiv \left| \frac{1}{2} \frac{-1}{2} \right\rangle = -\sqrt{2/3} \left| 1 \frac{1}{2} -1 \frac{1}{2} \right\rangle + \sqrt{1/3} \left| 1 \frac{1}{2} 0 \frac{-1}{2} \right\rangle.$$

In this basis the nonzero matrix elements of H'_{fs} are all on the diagonal, and given by Equation 7.68; H'_Z has four off-diagonal elements, and the complete matrix $-W$ is (see Problem 7.29):

$$\begin{pmatrix} 5\gamma - \beta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5\gamma + \beta & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma - 2\beta & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \gamma + 2\beta & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \gamma - \frac{2}{3}\beta & \frac{\sqrt{2}}{3}\beta & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{3}\beta & 5\gamma - \frac{1}{3}\beta & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \gamma + \frac{2}{3}\beta & \frac{\sqrt{2}}{3}\beta \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{3}\beta & 5\gamma + \frac{1}{3}\beta \end{pmatrix}$$

²⁷ You can use ℓ, m_ℓ, m_s states if you prefer—this makes the matrix elements of H'_Z easier, but those of H'_{fs} more difficult; the W matrix will be more complicated, but its eigenvalues (which are independent of basis) are the same either way.

²⁸ Don't confuse the notation $|\ell s m_\ell m_s\rangle$ in the Clebsch–Gordan tables with $|n \ell j m_j\rangle$ (in Section 7.4.1) or $|n \ell m_\ell m_s\rangle$ (in Section 7.4.2); here n is always 2, and s (of course) is always $1/2$.

Table 7.2: Energy levels for the $n = 2$ states of hydrogen, with fine structure and Zeeman splitting.

ϵ_1	$= E_2 - 5\gamma + \beta$
ϵ_2	$= E_2 - 5\gamma - \beta$
ϵ_3	$= E_2 - \gamma + 2\beta$
ϵ_4	$= E_2 - \gamma - 2\beta$
ϵ_5	$= E_2 - 3\gamma + \beta/2 + \sqrt{4\gamma^2 + (2/3)\gamma\beta + \beta^2/4}$
ϵ_6	$= E_2 - 3\gamma + \beta/2 - \sqrt{4\gamma^2 + (2/3)\gamma\beta + \beta^2/4}$
ϵ_7	$= E_2 - 3\gamma - \beta/2 + \sqrt{4\gamma^2 - (2/3)\gamma\beta + \beta^2/4}$
ϵ_8	$= E_2 - 3\gamma - \beta/2 - \sqrt{4\gamma^2 - (2/3)\gamma\beta + \beta^2/4}$

where

$$\gamma \equiv (\alpha/8)^2 \, 13.6 \text{ eV} \quad \text{and} \quad \beta \equiv \mu_B B_{\text{ext}}.$$

The first four eigenvalues are already displayed along the diagonal; it remains only to find the eigenvalues of the two 2×2 blocks. The characteristic equation for the first of these is

$$\lambda^2 - \lambda(6\gamma - \beta) + \left(5\gamma^2 - \frac{11}{3}\gamma\beta\right) = 0,$$

and the quadratic formula gives the eigenvalues:

$$\lambda_{\pm} = 3\gamma - (\beta/2) \pm \sqrt{4\gamma^2 + (2/3)\gamma\beta + (\beta^2/4)}. \quad (7.88)$$

The eigenvalues of the second block are the same, but with the sign of β reversed. The eight energies are listed in Table 7.2, and plotted against B_{ext} in Figure 7.11. In the zero-field limit ($\beta = 0$) they reduce to the fine structure values; for weak fields ($\beta \ll \gamma$) they reproduce what you got in Problem 7.24; for strong fields ($\beta \gg \gamma$) we recover the results of Problem 7.27 (note the convergence to five distinct energy levels, at very high fields, as predicted in Problem 7.27).

Problem 7.29 Work out the matrix elements of H'_Z and H'_{fs} , and construct the W matrix given in the text, for $n = 2$.

Problem 7.30 Analyze the Zeeman effect for the $n = 3$ states of hydrogen, in the weak, strong, and intermediate field regimes. Construct a table of energies (analogous to Table 7.2), plot them as functions of the external field (as in Figure 7.11), and check that the intermediate-field results reduce properly in the two limiting cases. *Hint:* The Wigner–Eckart theorem comes in handy here. In Chapter 6 we wrote the theorem in terms of the orbital angular momentum ℓ but it also holds for states of total angular momentum j . In particular,

$$\langle j'm'_j | V^z | jm_j \rangle = C_{mj0m'_j}^{j-1j} \langle j' \| V \| j \rangle$$

for any vector operator $\mathbf{V}$ (and $\mathbf{L} + 2\mathbf{S}$ is a vector operator).

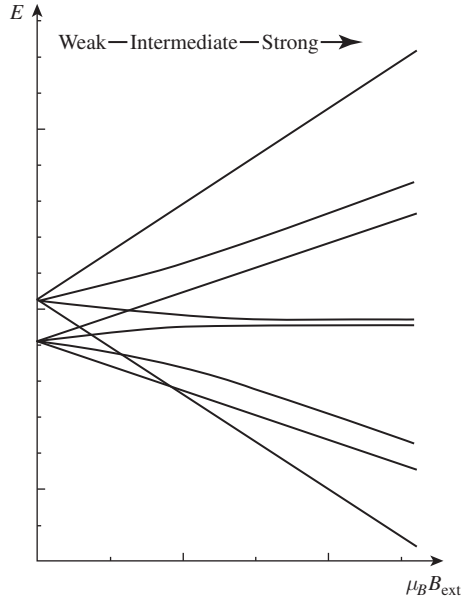


Figure 7.11: Zeeman splitting of the $n = 2$ states of hydrogen, in the weak, intermediate, and strong field regimes.

7.5 HYPERFINE SPLITTING IN HYDROGEN

The proton itself constitutes a magnetic dipole, though its dipole moment is much smaller than the electron's because of the mass in the denominator (Equation 7.62):

$$\boldsymbol{\mu}_p = \frac{g_p e}{2m_p} \mathbf{S}_p, \quad \boldsymbol{\mu}_e = -\frac{e}{m_e} \mathbf{S}_e. \quad (7.89)$$

(The proton is a composite structure, made up of three quarks, and its gyromagnetic ratio is not as simple as the electron's—hence the explicit g -factor (g_p), whose measured value is 5.59 as opposed to 2.00 for the electron.) According to classical electrodynamics, a dipole $\boldsymbol{\mu}$ sets up a magnetic field²⁹

$$\mathbf{B} = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \boldsymbol{\mu}] + \frac{2\mu_0}{3} \boldsymbol{\mu} \delta^3(\mathbf{r}). \quad (7.90)$$

So the Hamiltonian of the electron, in the magnetic field due to the proton's magnetic dipole moment, is (Equation 7.59)

$$H'_{\text{hf}} = \frac{\mu_0 g_p e^2}{8\pi m_p m_e} \frac{[3(\mathbf{S}_p \cdot \hat{\mathbf{r}})(\mathbf{S}_e \cdot \hat{\mathbf{r}}) - \mathbf{S}_p \cdot \mathbf{S}_e]}{r^3} + \frac{\mu_0 g_p e^2}{3m_p m_e} \mathbf{S}_p \cdot \mathbf{S}_e \delta^3(\mathbf{r}). \quad (7.91)$$

²⁹ If you are unfamiliar with the delta function term in Equation 7.90, you can derive it by treating the dipole as a spinning charged spherical shell, in the limit as the radius goes to zero and the charge goes to infinity (with $\boldsymbol{\mu}$ held constant). See D. J. Griffiths, *Am. J. Phys.*, **50**, 698 (1982).

According to perturbation theory, the first-order correction to the energy (Equation 7.9) is the expectation value of the perturbing Hamiltonian:

$$E_{\text{hf}}^1 = \frac{\mu_0 g_p e^2}{8\pi m_p m_e} \left\langle \frac{3 (\mathbf{S}_p \cdot \hat{r}) (\mathbf{S}_e \cdot \hat{r}) - \mathbf{S}_p \cdot \mathbf{S}_e}{r^3} \right\rangle + \frac{\mu_0 g_p e^2}{3m_p m_e} \langle \mathbf{S}_p \cdot \mathbf{S}_e \rangle |\psi(0)|^2. \quad (7.92)$$

In the ground state (or any other state for which $\ell = 0$) the wave function is spherically symmetric, and the first expectation value vanishes (see Problem 7.31). Meanwhile, from Equation 4.80 we find that $|\psi_{100}(0)|^2 = 1/(\pi a^3)$, so

$$E_{\text{hf}}^1 = \frac{\mu_0 g_p e^2}{3\pi m_p m_e a^3} \langle \mathbf{S}_p \cdot \mathbf{S}_e \rangle, \quad (7.93)$$

in the ground state. This is called **spin-spin coupling**, because it involves the dot product of two spins (contrast spin-orbit coupling, which involves $\mathbf{S} \cdot \mathbf{L}$).

In the presence of spin-spin coupling, the individual spin angular momenta are no longer conserved; the “good” states are eigenvectors of the *total* spin,

$$\mathbf{S} \equiv \mathbf{S}_e + \mathbf{S}_p. \quad (7.94)$$

As before, we square this out to get

$$\mathbf{S}_p \cdot \mathbf{S}_e = \frac{1}{2} (S^2 - S_e^2 - S_p^2). \quad (7.95)$$

But the electron and proton both have spin $1/2$, so $S_e^2 = S_p^2 = (3/4)\hbar^2$. In the triplet state (spins “parallel”) the total spin is 1, and hence $S^2 = 2\hbar^2$; in the singlet state the total spin is 0, and $S^2 = 0$. Thus

$$E_{\text{hf}}^1 = \frac{4g_p \hbar^4}{3m_p m_e^2 c^2 a^4} \begin{cases} +1/4, & \text{(triplet)}; \\ -3/4, & \text{(singlet)}. \end{cases} \quad (7.96)$$

Spin-spin coupling breaks the spin degeneracy of the ground state, lifting the triplet configuration and depressing the singlet (see Figure 7.12). The energy gap is

$$\Delta E = \frac{4g_p \hbar^4}{3m_p m_e^2 c^2 a^4} = 5.88 \times 10^{-6} \text{ eV}. \quad (7.97)$$

The frequency of the photon emitted in a transition from the triplet to the singlet state is

$$\nu = \frac{\Delta E}{h} = 1420 \text{ MHz}, \quad (7.98)$$

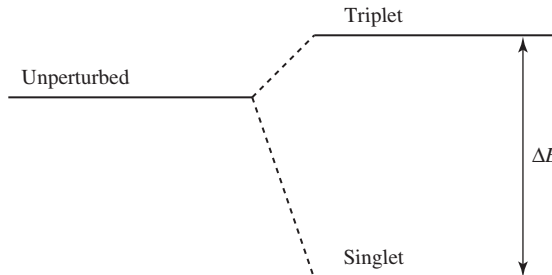


Figure 7.12: Hyperfine splitting in the ground state of hydrogen.

and the corresponding wavelength is $c/\nu = 21$ cm, which falls in the microwave region. This famous **21-centimeter line** is among the most pervasive forms of radiation in the universe.

Problem 7.31 Let $\mathbf{a}$ and $\mathbf{b}$ be two constant vectors. Show that

$$\int (\mathbf{a} \cdot \hat{\mathbf{r}}) (\mathbf{b} \cdot \hat{\mathbf{r}}) \sin \theta \, d\theta \, d\phi = \frac{4\pi}{3} (\mathbf{a} \cdot \mathbf{b}) \quad (7.99)$$

(the integration is over the usual range: $0 < \theta < \pi$, $0 < \phi < 2\pi$). Use this result to demonstrate that

$$\left\langle \frac{3 (\mathbf{S}_p \cdot \hat{\mathbf{r}}) (\mathbf{S}_e \cdot \hat{\mathbf{r}}) - \mathbf{S}_p \cdot \mathbf{S}_e}{r^3} \right\rangle = 0,$$

for states with $\ell = 0$. *Hint:* $\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{i}} + \sin \theta \sin \phi \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}}$. Do the angular integrals first.

Problem 7.32 By appropriate modification of the hydrogen formula, determine the hyperfine splitting in the ground state of (a) **muonic hydrogen** (in which a muon—same charge and g -factor as the electron, but 207 times the mass—substitutes for the electron), (b) **positronium** (in which a positron—same mass and g -factor as the electron, but opposite charge—substitutes for the proton), and (c) **muonium** (in which an anti-muon—same mass and g -factor as a muon, but opposite charge—substitutes for the proton). *Hint:* Don't forget to use the reduced mass (Problem 5.1) in calculating the “Bohr radius” of these exotic “atoms,” but use the *actual* masses in the gyromagnetic ratios. Incidentally, the answer you get for positronium (4.82×10^{-4} eV) is quite far from the experimental value (8.41×10^{-4} eV); the large discrepancy is due to pair annihilation ($e^+ + e^- \rightarrow \gamma + \gamma$), which contributes an extra $(3/4) \Delta E$, and does not occur (of course) in ordinary hydrogen, muonic hydrogen, or muonium.³⁰

FURTHER PROBLEMS ON CHAPTER 7

Problem 7.33 Estimate the correction to the ground state energy of hydrogen due to the finite size of the nucleus. Treat the proton as a uniformly charged spherical shell of radius b , so the potential energy of an electron inside the shell is *constant*: $-e^2/(4\pi\epsilon_0 b)$; this isn't very realistic, but it is the simplest model, and it will give us the right order of magnitude. Expand your result in powers of the small parameter (b/a) , where a is the Bohr radius, and keep only the leading term, so your final answer takes the form

$$\frac{\Delta E}{E} = A (b/a)^n.$$

Your business is to determine the constant A and the power n . Finally, put in $b \approx 10^{-15}$ m (roughly the radius of the proton) and work out the actual number. How does it compare with fine structure and hyperfine structure?

³⁰ For details see Griffiths, footnote 29, page 311.

Problem 7.34 In this problem you will develop an alternative approach to degenerate perturbation theory. Consider an unperturbed Hamiltonian H^0 with two degenerate states ψ_a^0 and ψ_b^0 (energy E^0), and a perturbation H' . Define the operator that projects³¹ onto the degenerate subspace:

$$P_D = |\psi_a^0\rangle\langle\psi_a^0| + |\psi_b^0\rangle\langle\psi_b^0|. \quad (7.100)$$

The Hamiltonian can be written

$$H = H^0 + H' = \tilde{H}^0 + \tilde{H}' \quad (7.101)$$

where

$$\tilde{H}^0 = H^0 + P_D H' P_D, \quad \tilde{H}' = H' - P_D H' P_D. \quad (7.102)$$

The idea is to treat $\tilde{H}^0$ as the “unperturbed” Hamiltonian and $\tilde{H}'$ as the perturbation; as you’ll soon discover, $\tilde{H}^0$ is nondegenerate, so we can use ordinary nondegenerate perturbation theory.

(a) First we need to find the eigenstates of $\tilde{H}^0$.

- i. Show that any eigenstate ψ_n^0 (other than ψ_a^0 or ψ_b^0) of H^0 is also an eigenstate of $\tilde{H}^0$ with the same eigenvalue.
- ii. Show that the “good” states $\psi^0 = \alpha \psi_a^0 + \beta \psi_b^0$ (with α and β determined by solving Equation 7.30) are eigenstates of $\tilde{H}^0$ with energies $E^0 + E_{\pm}^1$.

(b) Assuming that E_+^1 and E_-^1 are distinct, you now have a *nondegenerate* unperturbed Hamiltonian $\tilde{H}^0$ and you can do nondegenerate perturbation theory using the perturbation $\tilde{H}'$. Find an expression for the energy to second order for the states $\psi_{\pm}^0$ in (ii).

Comment: One advantage of this approach is that it also handles the case where the unperturbed energies are not *exactly* equal, but very close:³² $E_a^0 \approx E_b^0$. In this case one must still use degenerate perturbation theory; an important example of this occurs in the **nearly-free electron approximation** for calculating band structure.³³

Problem 7.35 Here is an application of the technique developed in Problem 7.34. Consider the Hamiltonian

$$H^0 = \begin{pmatrix} \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon' \end{pmatrix}, \quad H' = \begin{pmatrix} 0 & a & b \\ a^* & 0 & c \\ b^* & c^* & 0 \end{pmatrix}.$$

(a) Find the projection operator P_D (it’s a 3×3 matrix) that projects onto the subspace spanned by

$$|\psi_a^0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad |\psi_b^0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Then construct the matrices $\tilde{H}^0$ and $\tilde{H}'$.

³¹ See page 118 for a discussion of projection operators.

³² See Problem 7.4 for a discussion of what close means in this context.

³³ See, for example, Steven H. Simon, *The Oxford Solid State Basics* (Oxford University Press, 2013), Section 15.1.

- (b) Solve for the eigenstates of $\tilde{H}^0$ and verify...
- that its spectrum is nondegenerate,
 - that the nondegenerate eigenstate of H^0

$$|\psi_c^0\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

is also an eigenstate of $\tilde{H}^0$ with the same eigenvalue.

- (c) What are the “good” states, and what are their energies, to first order in the perturbation?

Problem 7.36 Consider the isotropic three-dimensional harmonic oscillator (Problem 4.46). Discuss the effect (in first order) of the perturbation

$$H' = \lambda x^2 yz$$

(for some constant λ) on

- the ground state;
- the (triply degenerate) first excited state. *Hint:* Use the answers to Problems 2.12 and 3.39.

Problem 7.37 Van der Waals interaction. Consider two atoms a distance R apart. Because they are electrically neutral you might suppose there would be no force between them, but if they are polarizable there is in fact a weak attraction. To model this system, picture each atom as an electron (mass m , charge $-e$) attached by a spring (spring constant k) to the nucleus (charge $+e$), as in Figure 7.13. We'll assume the nuclei are heavy, and essentially motionless. The Hamiltonian for the unperturbed system is

$$H^0 = \frac{1}{2m} p_1^2 + \frac{1}{2} k x_1^2 + \frac{1}{2m} p_2^2 + \frac{1}{2} k x_2^2. \quad (7.103)$$

The Coulomb interaction between the atoms is

$$H' = \frac{1}{4\pi\epsilon_0} \left(\frac{e^2}{R} - \frac{e^2}{R-x_1} - \frac{e^2}{R+x_2} + \frac{e^2}{R-x_1+x_2} \right). \quad (7.104)$$

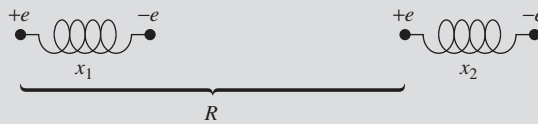


Figure 7.13: Two nearby polarizable atoms (Problem 7.37).

- Explain Equation 7.104. Assuming that $|x_1|$ and $|x_2|$ are both much less than R , show that

$$H' \cong -\frac{e^2 x_1 x_2}{2\pi\epsilon_0 R^3}. \quad (7.105)$$

- Show that the total Hamiltonian (H^0 plus Equation 7.105) separates into two harmonic oscillator Hamiltonians:

$$H = \left[\frac{1}{2m} p_+^2 + \frac{1}{2} \left(k - \frac{e^2}{2\pi\epsilon_0 R^3} \right) x_+^2 \right] + \left[\frac{1}{2m} p_-^2 + \frac{1}{2} \left(k + \frac{e^2}{2\pi\epsilon_0 R^3} \right) x_-^2 \right], \quad (7.106)$$

under the change variables

$$x_{\pm} \equiv \frac{1}{\sqrt{2}} (x_1 \pm x_2), \quad \text{which entails} \quad p_{\pm} = \frac{1}{\sqrt{2}} (p_1 \pm p_2). \quad (7.107)$$

(c) The ground state energy for this Hamiltonian is evidently

$$E = \frac{1}{2} \hbar (\omega_+ + \omega_-), \quad \text{where} \quad \omega_{\pm} = \sqrt{\frac{k \mp (e^2/2\pi\epsilon_0 R^3)}{m}}. \quad (7.108)$$

Without the Coulomb interaction it would have been $E_0 = \hbar\omega_0$, where $\omega_0 = \sqrt{k/m}$. Assuming that $k \gg (e^2/2\pi\epsilon_0 R^3)$, show that

$$\Delta V \equiv E - E_0 \cong -\frac{\hbar}{8m^2\omega_0^3} \left(\frac{e^2}{2\pi\epsilon_0} \right)^2 \frac{1}{R^6}. \quad (7.109)$$

Conclusion: There is an attractive potential between the atoms, proportional to the inverse sixth power of their separation. This is the **van der Waals interaction** between two neutral atoms.

(d) Now do the same calculation using second-order perturbation theory. *Hint:* The unperturbed states are of the form $\psi_{n_1}(x_1) \psi_{n_2}(x_2)$, where $\psi_n(x)$ is a one-particle oscillator wave function with mass m and spring constant k ; ΔV is the second-order correction to the ground state energy, for the perturbation in Equation 7.105 (notice that the *first-order* correction is zero).³⁴

**

Problem 7.38 Suppose the Hamiltonian H , for a particular quantum system, is a function of some parameter λ ; let $E_n(\lambda)$ and $\psi_n(\lambda)$ be the eigenvalues and eigenfunctions of $H(\lambda)$. The **Feynman–Hellmann theorem**³⁵ states that

$$\frac{\partial E_n}{\partial \lambda} = \left\langle \psi_n \left| \frac{\partial H}{\partial \lambda} \right| \psi_n \right\rangle \quad (7.110)$$

(assuming either that E_n is nondegenerate, or—if degenerate—that the ψ_n s are the “good” linear combinations of the degenerate eigenfunctions).

(a) Prove the Feynman–Hellmann theorem. *Hint:* Use Equation 7.9.

(b) Apply it to the one-dimensional harmonic oscillator, (i) using $\lambda = \omega$ (this yields a formula for the expectation value of V), (ii) using $\lambda = \hbar$ (this yields $\langle T \rangle$), and (iii) using $\lambda = m$ (this yields a relation between $\langle T \rangle$ and $\langle V \rangle$). Compare your answers to Problem 2.12, and the virial theorem predictions (Problem 3.37).

³⁴ There is an interesting fraud in this well-known problem. If you expand H' to order $1/R^5$, the extra term has a nonzero expectation value in the ground state of H^0 , so there is a nonzero first-order perturbation, and the dominant contribution goes like $1/R^5$, not $1/R^6$. The model gets the power “right” in three dimensions (where the expectation value is zero), but not in one. See A. C. Ipsen and K. Splittorff, *Am. J. Phys.* **83**, 150 (2015).

³⁵ Feynman obtained Equation 7.110 while working on his undergraduate thesis at MIT (R. P. Feynman, *Phys. Rev.* **56**, 340, 1939); Hellmann’s work was published four years earlier in an obscure Russian journal.

Problem 7.39 Consider a three-level system with the unperturbed Hamiltonian

$$H^0 = \begin{pmatrix} \epsilon_a & 0 & 0 \\ 0 & \epsilon_a & 0 \\ 0 & 0 & \epsilon_c \end{pmatrix} \quad (7.111)$$

($\epsilon_a > \epsilon_c$) and the perturbation

$$H' = \begin{pmatrix} 0 & 0 & V \\ 0 & 0 & V \\ V^* & V^* & 0 \end{pmatrix}. \quad (7.112)$$

Since the (2×2) matrix W is diagonal (and in fact identically 0) in the basis of states $(1, 0, 0)$ and $(0, 1, 0)$, you might assume they are the good states, but they're not. To see this:

- (a) Obtain the exact eigenvalues for the perturbed Hamiltonian $H = H^0 + H'$.
- (b) Expand your results from part (a) as a power series in $|V|$ up to second order.
- (c) What do you obtain by applying *nondegenerate* perturbation theory to find the energies of all three states (up to second order)? This would work if the assumption about the good states above were correct.

Moral: If any of the eigenvalues of W are equal, the states that diagonalize W are not unique, and diagonalizing W does not determine the “good” states. When this happens (and it's not uncommon), you need to use *second-order degenerate* perturbation theory (see Problem 7.40).

Problem 7.40 If it happens that the square root in Equation 7.33 vanishes, then $E_+^1 = E_-^1$; the degeneracy is not lifted at first order. In this case, diagonalizing the W matrix puts no restriction on α and β and you still don't know what the “good” states are. If you need to determine the “good” states—for example to calculate higher-order corrections—you need to use **second-order degenerate perturbation theory**.

- (a) Show that, for the two-fold degeneracy studied in Section 7.2.1, the first-order correction to the wave function in degenerate perturbation theory is

$$\psi^1 = \sum_{m \neq a, b} \frac{\alpha V_{ma} + \beta V_{mb}}{E^0 - E_m^0} \psi_m^0.$$

- (b) Consider the terms of order λ^2 (corresponding to Equation 7.8 in the nondegenerate case) to show that α and β are determined by finding the eigenvectors of the matrix W^2 (the superscript denotes second order, not W squared) where

$$[W^2]_{ij} = \sum_{m \neq a, b} \frac{\langle \psi_i^0 | H' | \psi_m^0 \rangle \langle \psi_m^0 | H' | \psi_j^0 \rangle}{E^0 - E_m^0}$$

and that the eigenvalues of this matrix correspond to the second-order energies E^2 .

- (c) Show that second-order degenerate perturbation theory, developed in (b), gives the correct energies to second order for the three-state Hamiltonian in Problem 7.39.

**

Problem 7.41 A free particle of mass m is confined to a ring of circumference L such that $\psi(x + L) = \psi(x)$. The unperturbed Hamiltonian is

$$H^0 = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2},$$

to which we add a perturbation

$$H' = V_0 \cos\left(2\pi \frac{x}{L}\right).$$

(a) Show that the unperturbed states may be written

$$\psi_n^0(x) = \frac{1}{\sqrt{L}} e^{i 2\pi n x/L}$$

for $n = 0, \pm 1, \pm 2$ and that, apart from $n = 0$, all of these states are two-fold degenerate.

(b) Find a general expression for the matrix elements of the perturbation:

$$H'_{mn} = \langle \psi_m^0 | H' | \psi_n^0 \rangle.$$

(c) Consider the degenerate pair of states with $n = \pm 1$. Construct the matrix W and calculate the first-order energy corrections, E^1 . Note that the degeneracy *does not lift at first order*. Therefore, diagonalizing W does not tell us what the “good” states are.

(d) Construct the matrix W^2 (Problem 7.40) for the states $n = \pm 1$, and show that the degeneracy lifts at second order. What are the good linear combinations of the states with $n = \pm 1$?

(e) What are the energies, accurate to second order, for these states?³⁶

**

Problem 7.42 The Feynman–Hellmann theorem (Problem 7.38) can be used to determine the expectation values of $1/r$ and $1/r^2$ for hydrogen.³⁷ The effective Hamiltonian for the radial wave functions is (Equation 4.53)

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{\hbar^2}{2m} \frac{\ell(\ell+1)}{r^2} - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r},$$

and the eigenvalues (expressed in terms of ℓ)³⁸ are (Equation 4.70)

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2 (N + \ell)^2}.$$

(a) Use $\lambda = e$ in the Feynman–Hellmann theorem to obtain $\langle 1/r \rangle$. Check your result against Equation 7.56.

(b) Use $\lambda = \ell$ to obtain $\langle 1/r^2 \rangle$. Check your answer with Equation 7.57.

³⁶ See D. Kiang, *Am. J. Phys.* **46** (11), 1978 and L.-K. Chen, *Am. J. Phys.* **72** (7), 2004 for further discussion of this problem. It is shown that each degenerate energy level, $E_{\pm n}^0$, splits at order $2n$ in perturbation theory. The exact solution to the problem can also be obtained as the time-independent Schrödinger equation for $H = H^0 + H^1$ reduces to the **Mathieu equation**.

³⁷ C. Sánchez del Río, *Am. J. Phys.*, **50**, 556 (1982); H. S. Valk, *Am. J. Phys.*, **54**, 921 (1986).

³⁸ In part (b) we treat ℓ as a continuous variable; n becomes a function of ℓ , according to Equation 4.67, because N , which must be an integer, is fixed. To avoid confusion, I have eliminated n , to reveal the dependence on ℓ explicitly.

Problem 7.43 Prove **Kramers' relation**.³⁹

$$\frac{s+1}{n^2} \langle r^s \rangle - (2s+1) a \langle r^{s-1} \rangle + \frac{s}{4} \left[(2\ell+1)^2 - s^2 \right] a^2 \langle r^{s-2} \rangle = 0, \quad (7.113)$$

which relates the expectation values of r to three different powers (s , $s-1$, and $s-2$), for an electron in the state $\psi_{n\ell m}$ of hydrogen. *Hint:* Rewrite the radial equation (Equation 4.53) in the form

$$u'' = \left[\frac{\ell(\ell+1)}{r^2} - \frac{2}{ar} + \frac{1}{n^2 a^2} \right] u,$$

and use it to express $\int (ur^s u'') dr$ in terms of $\langle r^s \rangle$, $\langle r^{s-1} \rangle$, and $\langle r^{s-2} \rangle$. Then use integration by parts to reduce the second derivative. Show that $\int (ur^s u') dr = -(s/2) \langle r^{s-1} \rangle$, and $\int (u' r^s u') dr = -[2/(s+1)] \int (u'' r^{s+1} u') dr$. Take it from there.

Problem 7.44

- (a) Plug $s = 0$, $s = 1$, $s = 2$, and $s = 3$ into Kramers' relation (Equation 7.113) to obtain formulas for $\langle r^{-1} \rangle$, $\langle r \rangle$, $\langle r^2 \rangle$, and $\langle r^3 \rangle$. Note that you could continue indefinitely, to find *any* positive power.
- (b) In the *other* direction, however, you hit a snag. Put in $s = -1$, and show that all you get is a relation between $\langle r^{-2} \rangle$ and $\langle r^{-3} \rangle$.
- (c) But if you can get $\langle r^{-2} \rangle$ by some *other* means, you can apply the Kramers relation to obtain the rest of the negative powers. Use Equation 7.57 (which is derived in Problem 7.42) to determine $\langle r^{-3} \rangle$, and check your answer against Equation 7.66.

Problem 7.45 When an atom is placed in a uniform external electric field $\mathbf{E}_{\text{ext}}$, the energy levels are shifted—a phenomenon known as the **Stark effect** (it is the electrical analog to the Zeeman effect). In this problem we analyze the Stark effect for the $n = 1$ and $n = 2$ states of hydrogen. Let the field point in the z direction, so the potential energy of the electron is

$$H'_S = eE_{\text{ext}}z = eE_{\text{ext}}r \cos \theta.$$

Treat this as a perturbation on the Bohr Hamiltonian (Equation 7.43). (Spin is irrelevant to this problem, so ignore it, and neglect the fine structure.)

- (a) Show that the ground state energy is not affected by this perturbation, in first order.
- (b) The first excited state is four-fold degenerate: ψ_{200} , ψ_{211} , ψ_{210} , ψ_{21-1} . Using degenerate perturbation theory, determine the first-order corrections to the energy. Into how many levels does E_2 split?
- (c) What are the “good” wave functions for part (b)? Find the expectation value of the electric dipole moment ($\mathbf{p}_e = -e\mathbf{r}$), in each of these “good” states. Notice that the results are independent of the applied field—evidently hydrogen in its first excited state can carry a *permanent* electric dipole moment.

³⁹ This is also known as the (second) **Pasternack relation**. See H. Beker, *Am. J. Phys.* **65**, 1118 (1997). For a proof based on the Feynman–Hellmann theorem (Problem 7.38) see S. Balasubramanian, *Am. J. Phys.* **68**, 959 (2000).

Hint: There are lots of integrals in this problem, but almost all of them are zero. So study each one carefully, before you do any calculations: If the ϕ integral vanishes, there's not much point in doing the r and θ integrals! You can avoid those integrals altogether if you use the selection rules of Sections 6.4.3 and 6.7.2. *Partial answer:* $W_{13} = W_{31} = -3eaE_{\text{ext}}$; all other elements are zero.

Problem 7.46 Consider the Stark effect (Problem 7.45) for the $n = 3$ states of hydrogen. There are initially nine degenerate states, $\psi_{3\ell m}$ (neglecting spin, as before), and we turn on an electric field in the z direction.

(a) Construct the 9×9 matrix representing the perturbing Hamiltonian. *Partial answer:* $\langle 300|z|310\rangle = -3\sqrt{6}a$, $\langle 310|z|320\rangle = -3\sqrt{3}a$, $\langle 31\pm 1|z|32\pm 1\rangle = -(9/2)a$.

(b) Find the eigenvalues, and their degeneracies.

Problem 7.47 Calculate the wavelength, in centimeters, of the photon emitted under a hyperfine transition in the ground state ($n = 1$) of **deuterium**. Deuterium is “heavy” hydrogen, with an extra neutron in the nucleus; the proton and neutron bind together to form a **deuteron**, with spin 1 and magnetic moment

$$\mu_d = \frac{gde}{2m_d} \mathbf{S}_d;$$

the deuteron g -factor is 1.71.

Problem 7.48 In a crystal, the electric field of neighboring ions perturbs the energy levels of an atom. As a crude model, imagine that a hydrogen atom is surrounded by three pairs of point charges, as shown in Figure 7.14. (Spin is irrelevant to this problem, so ignore it.)

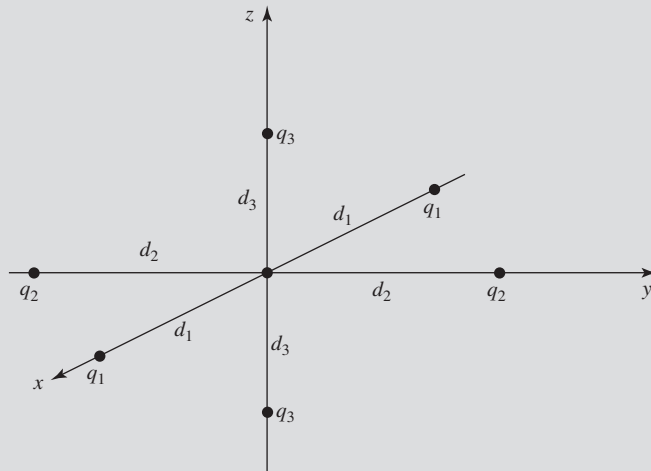


Figure 7.14: Hydrogen atom surrounded by six point charges (crude model for a crystal lattice); Problem 7.48.

(a) Assuming that $r \ll d_1$, $r \ll d_2$, and $r \ll d_3$, show that

$$H' = V_0 + 3 \left(\beta_1 x^2 + \beta_2 y^2 + \beta_3 z^2 \right) - (\beta_1 + \beta_2 + \beta_3) r^2,$$

where

$$\beta_i \equiv -\frac{e}{4\pi\epsilon_0} \frac{q_i}{d_i^3}, \quad \text{and } V_0 = 2 \left(\beta_1 d_1^2 + \beta_2 d_2^2 + \beta_3 d_3^2 \right).$$

(b) Find the lowest-order correction to the ground state energy.

(c) Calculate the first-order corrections to the energy of the first excited states ($n = 2$). Into how many levels does this four-fold degenerate system split, (i) in the case of **cubic symmetry**, $\beta_1 = \beta_2 = \beta_3$; (ii) in the case of **tetragonal symmetry**, $\beta_1 = \beta_2 \neq \beta_3$; (iii) in the general case of **orthorhombic symmetry** (all three different)? *Note:* you might recognize the “good” states from Problem 4.71.

Problem 7.49 A hydrogen atom is placed in a uniform magnetic field $\mathbf{B}_0 = B_0 \hat{z}$ (the Hamiltonian can be written as in Equation 4.230). Use the Feynman–Hellman theorem (Problem 7.38) to show that

$$\frac{\partial E_n}{\partial B_0} = -\langle \psi_n | \mu_z | \psi_n \rangle \quad (7.114)$$

where the electron’s magnetic dipole moment⁴⁰ (orbital plus spin) is

$$\boldsymbol{\mu} = \gamma_o \mathbf{L}_{\text{mechanical}} + \gamma \mathbf{S}.$$

The mechanical angular momentum is defined in Equation 4.231.

Note: From Equation 7.114 it follows that the magnetic susceptibility of N atoms in a volume V and at 0 K (when they’re all in the ground state) is⁴¹

$$\chi = \mu_0 \frac{\partial M}{\partial B_0} = -\frac{N}{V} \mu_0 \frac{\partial^2 E_0}{\partial B_0^2}, \quad (7.115)$$

where E_0 is the ground-state energy. Although we derived Equation 7.114 for a hydrogen atom, the expression applies to multi-electron atoms as well—even when electron–electron interactions are included.

Problem 7.50 For an atom in a uniform magnetic field $\mathbf{B}_0 = B_0 \hat{z}$, Equation 4.230 gives

$$H = H_{\text{atom}} - B_0 (\gamma_o L_z + \gamma S_z) + \frac{e^2}{8m} B_0^2 \sum_{i=1}^Z (x_i^2 + y_i^2),$$

where L_z and S_z refer to the total orbital and spin angular momentum of all the electrons.

⁴⁰ For most purposes we can take this to be the magnetic moment of the atom as well. The proton’s larger mass means that its contribution to the dipole moment is orders of magnitude smaller than the electron’s contribution.

⁴¹ See Problem 5.33 for the definition of magnetic susceptibility. This formula does not apply when the ground state is degenerate (see Neil W. Ashcroft and N. David Mermin, *Solid State Physics* (Belmont: Cengage, 1976), p. 655); atoms with non-degenerate ground states have $J = 0$ (see Table 5.1).

- (a) Treating the terms involving B_0 as a perturbation, compute the shift of the ground state energy of a helium atom to second order in B_0 . Assume that the helium ground state is given by

$$\psi_0 = \psi_{100}(r_1) \psi_{100}(r_2) \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}$$

where ψ_{100} refers to the hydrogenic ground state (with $Z = 2$).

- (b) Use the results of Problem 7.49 to calculate the magnetic susceptibility of helium. Given a density of 0.166 kg/m^3 , obtain a numerical value for the susceptibility. *Note:* The experimental result is -1.0×10^{-9} (the negative sign means that helium is a **diamagnet**). The results can be brought closer by taking account of screening, which increases the orbital radius (see Section 8.2).

Problem 7.51 Sometimes it is possible to solve Equation 7.10 directly, without having to expand ψ_n^1 in terms of the unperturbed wave functions (Equation 7.11). Here are two particularly nice examples.

- (a) **Stark effect in the ground state of hydrogen.**

- (i) Find the first-order correction to the ground state of hydrogen in the presence of a uniform external electric field E_{ext} (see Problem 7.45). *Hint:* Try a solution of the form

$$(A + Br + Cr^2)e^{-r/a} \cos \theta;$$

your problem is to find the constants A , B , and C that solve Equation 7.10.

- (ii) Use Equation 7.14 to determine the second-order correction to the ground state energy (the first-order correction is zero, as you found in Problem 7.45(a)). *Answer:* $-m(3a^2 e E_{\text{ext}}/2\hbar)^2$.

- (b) If the proton had an *electric* dipole moment p , the potential energy of the electron in hydrogen would be perturbed in the amount

$$H' = -\frac{ep \cos \theta}{4\pi\epsilon_0 r^2}.$$

- (i) Solve Equation 7.10 for the first-order correction to the ground state wave function.
(ii) Show that the *total* electric dipole moment of the atom is (surprisingly) *zero*, to this order.
(iii) Use Equation 7.14 to determine the second-order correction to the ground state energy. What is the *first-order* correction?

Problem 7.52 Consider a spinless particle of charge q and mass m constrained to move in the xy plane under the influence of the two-dimensional harmonic oscillator potential

$$V(x, y) = \frac{1}{2}m\omega^2(x^2 + y^2).$$

- (a) Construct the ground state wave function, $\psi_0(x, y)$, and write down its energy. Do the same for the (degenerate) first excited states.
(b) Now imagine that we turn on a weak magnetic field of magnitude B_0 pointing in the z -direction, so that (to first order in B_0) the Hamiltonian acquires an extra term

$$H' = -\boldsymbol{\mu} \cdot \mathbf{B} = -\frac{q}{2m} (\mathbf{L} \cdot \mathbf{B}) = -\frac{qB_0}{2m} (x p_y - y p_x).$$

Treating this as a perturbation, find the first-order corrections to the energies of the ground state and first excited states.



Problem 7.53 Imagine an infinite square well (Equation 2.22) into which we introduce a delta-function perturbation,

$$H'(x) = \lambda \delta(x - x_0),$$

where λ is a positive constant, and $0 < x_0 < a$ (to simplify matters, let $x_0 = pa$, where $0 < p < 1$).⁴²

- (a) Find the first-order correction to the n th allowed energy (Equation 2.30), assuming λ is small. (What does “small” mean, in this context?)
- (b) Find the second-order correction to the allowed energies. (Leave your answer as a sum.)
- (c) Now solve the Schrödinger equation exactly, treating separately the regions $0 \leq x < x_0$ and $x_0 < x < a$, and imposing the boundary conditions at x_0 . Derive the transcendental equation for the energies:

$$u_n \sin(u_n) + \Lambda \sin(pu_n) \sin[(1-p)u_n] = 0 \quad (E > 0). \quad (7.116)$$

Here $\Lambda \equiv 2ma\lambda/\hbar^2$, $u_n \equiv k_n a$, and $k_n \equiv \sqrt{2mE_n}/\hbar$. Check that Equation 7.116 reproduces your result from part (a), in the appropriate limit.

- (d) Everything so far holds just as well if λ is *negative*, but in that case there may be an additional solution with negative energy. Derive the transcendental equation for a negative-energy state:

$$v \sinh(v) + \Lambda \sinh(pv) \sinh[(1-p)v] = 0 \quad (E < 0), \quad (7.117)$$

where $v \equiv \kappa a$ and $\kappa \equiv \sqrt{-2mE}/\hbar$. Specialize to the symmetrical case $p = 1/2$, and show that you recover the energy of the delta-function well (Equation 2.132), in the appropriate regime.

- (e) There is in fact exactly *one* negative-energy solution, provided that $|\Lambda| > 1/[p(1-p)]$. First, prove this (graphically), for the case $p = 1/2$. (Below that critical value there is no negative-energy solution.) Next, by computer, plot the solution v , as a function of p , for $\Lambda = -4.1, -5$, and -10 . Verify that the solution only exists within the predicted range of p .
- (f) For $p = 1/2$, plot the ground state wave function, $\psi_1(x)$, for $\Lambda = 0, -2, -3, -3.5, -4.5, -5$, and -10 , to show how the sinusoidal shape (Figure 2.2) evolves into the exponential shape (Figure 2.13), as the delta function well “deepens.”⁴³

*** **Problem 7.54** Suppose you want to calculate the expectation value of some observable Ω , in the n th energy eigenstate of a system that is perturbed by H' :

$$\langle \Omega \rangle = \langle \psi_n | \hat{\Omega} | \psi_n \rangle.$$

⁴² We adopt the notation of Y. N. Joglekar, *Am. J. Phys.* **77**, 734 (2009), from which this problem is drawn.

⁴³ For the corresponding analysis of the delta function *barrier* (positive λ) see Problem 11.34.

Replacing ψ_n by its perturbation expansion, Equation 7.5,⁴⁴

$$\langle \Omega \rangle = \langle \psi_n^0 | \hat{\Omega} | \psi_n^0 \rangle + \lambda \left[\langle \psi_n^1 | \hat{\Omega} | \psi_n^0 \rangle + \langle \psi_n^0 | \hat{\Omega} | \psi_n^1 \rangle \right] + \lambda^2 (\dots) + \dots$$

The first-order correction to $\langle \Omega \rangle$ is therefore

$$\langle \Omega \rangle^1 = 2\text{Re} \left[\langle \psi_n^0 | \hat{\Omega} | \psi_n^1 \rangle \right],$$

or, using Equation 7.13,

$$\langle \Omega \rangle^1 = 2\text{Re} \sum_{m \neq n} \frac{\langle \psi_n^0 | \hat{\Omega} | \psi_m^0 \rangle \langle \psi_m^0 | H' | \psi_n^0 \rangle}{E_n^0 - E_m^0} \quad (7.118)$$

(assuming the unperturbed energies are nondegenerate, or that we are using the “good” basis states).

- (a) Suppose $\Omega = H'$ (the perturbation itself). What does Equation 7.118 tell us in this case? Explain (carefully) why this is consistent with Equation 7.15.
- (b) Consider a particle of charge q (maybe an electron in a hydrogen atom, or a pith ball connected to a spring), that is placed in a weak electric field E_{ext} pointing in the x direction, so that

$$H' = -qE_{\text{ext}}x.$$

The field will induce an electric dipole moment, $p_e = qx$, in the “atom.” The expectation value of p_e is proportional to the applied field, and the proportionality factor is called the **polarizability**, α . Show that

$$\alpha = -2q^2 \sum_{m \neq n} \frac{|\langle \psi_n^0 | x | \psi_m^0 \rangle|^2}{E_n^0 - E_m^0}. \quad (7.119)$$

Find the polarizability of the ground state of a one-dimensional harmonic oscillator. Compare the classical answer.

- (c) Now imagine a particle of mass m in a one-dimensional harmonic oscillator with a small anharmonic perturbation⁴⁵

$$H' = -\frac{1}{6}\kappa x^3. \quad (7.120)$$

Find $\langle x \rangle$ (to first order), in the n th energy eigenstate. *Answer:* $(n + \frac{1}{2}) \hbar \kappa / (2m^2 \omega^3)$.

Comment: As the temperature increases, higher-energy states are populated, and the particles move farther (on average) from their equilibrium positions; that’s why most solids expand with rising temperature.

Problem 7.55 Crandall’s Puzzle.⁴⁶ Stationary states of the one-dimensional Schrödinger equation ordinarily respect three “rules of thumb”: (1) the energies are nondegenerate,

⁴⁴ In general, Equation 7.5 does not deliver a *normalized* wave function, but the choice $c_n^{(n)} = 0$ in Equation 7.11 guarantees normalization to first order in λ , which is all we require here (see footnote 4, page 282).

⁴⁵ This is just a generic tweak to the simple harmonic oscillator potential, $\frac{1}{2}kx^2$; κ is some constant, and the factor of $-1/6$ is for convenience.

⁴⁶ Richard Crandall introduced me to this problem.

(2) the ground state has no nodes, the first excited state has one node, the second has two, and so on, and (3) if the potential is an even function of x , the ground state is even, the first excited state is odd, the second is even, and so on. We have already seen that the “bead-on-a-ring” (Problem 2.46) violates the first of these; now suppose we introduce a “nick” in at the origin:

$$H' = -\alpha\delta(x).$$

(If you don’t like the delta function, make it a gaussian, as in Problem 7.9.) This lifts the degeneracy, but what is the sequence of even and odd wave functions, and what is the sequence of node numbers? *Hint*: You don’t really need to do any calculations, here, and you’re welcome to assume that α is small, but by all means solve the Schrödinger equation exactly if you prefer.

Problem 7.56 In this problem we treat the electron–electron repulsion term in the helium Hamiltonian (Equation 5.38) as a perturbation,

$$H' = \frac{1}{4\pi\epsilon_0} \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|}.$$

(This will not be very accurate, because the perturbation is not small, in comparison to the Coulomb attraction of the nucleus ... but it’s a start.)

(a) Find the first-order correction to the ground state,

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2) = \psi_{100}(\mathbf{r}_1)\psi_{100}(\mathbf{r}_2).$$

(You have already done this calculation, if you worked Problem 5.15—only we didn’t call it perturbation theory back then.)

(b) Now treat the first excited state, in which one electron is in the hydrogenic ground state, ψ_{100} , and the other is in the state ψ_{200} . Actually, there are *two* such states, depending on whether the electron spins occupy the singlet configuration (parahelium) or the triplet (orthohelium):⁴⁷

$$\psi_{\pm}(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} [\psi_{100}(\mathbf{r}_1)\psi_{200}(\mathbf{r}_2) \pm \psi_{200}(\mathbf{r}_1)\psi_{100}(\mathbf{r}_2)].$$

Show that

$$E_{\pm}^1 = \frac{1}{2}(K \pm J),$$

where

$$K \equiv 2 \int \psi_{100}(\mathbf{r}_1)\psi_{200}(\mathbf{r}_2)H'\psi_{100}(\mathbf{r}_1)\psi_{200}(\mathbf{r}_2) d^3\mathbf{r}_1 d^3\mathbf{r}_2,$$

$$J \equiv 2 \int \psi_{100}(\mathbf{r}_1)\psi_{200}(\mathbf{r}_2)H'\psi_{200}(\mathbf{r}_1)\psi_{100}(\mathbf{r}_2) d^3\mathbf{r}_1 d^3\mathbf{r}_2.$$

Evaluate these two integrals, put in the actual numbers, and compare your results with Figure 5.2 (the measured energies are -59.2 eV and -58.4 eV).⁴⁸

⁴⁷ It seems strange, at first glance, that spin has anything to do with it, since the perturbation itself doesn’t involve spin (and I’m not even bothering to include the spin state explicitly). The point, of course, is that an antisymmetric spin state forces a symmetric (position) wave function, and vice versa, and this *does* affect the result.

⁴⁸ If you want to pursue this problem further, see R. C. Massé and T. G. Walker, *Am. J. Phys.* **83**, 730 (2015).

Problem 7.57 The Hamiltonian for the Bloch functions (Equation 6.12) can be analyzed with perturbation theory by defining H^0 and H' such that

$$\begin{aligned} H^0 u_{n0} &= E_{n0} u_{n0}, \\ (H^0 + H') u_{nq} &= E_{nq} u_{nq}. \end{aligned}$$

In this problem, don't assume anything about the form of $V(x)$.

- (a) Determine the operators H^0 and H' (express them in terms of $\hat{p}$).
 (b) Find E_{nq} to second order in q . That is, find expressions for A_n , B_n , and C_n (in terms of the E_{n0} and matrix elements of $\hat{p}$ in the “unperturbed” states u_{n0})

$$E_{nq} \approx A_n + B_n q + C_n q^2.$$

- (c) Show that the constants B_n are all zero. *Hint:* See Problem 2.1(b) to get started. Remember that $u_{n0}(x)$ is periodic.

Comment: It is conventional to write $C_n = \hbar^2/2m_n^*$ where m_n^* is the **effective mass** of particles in the n th band since then, as you've just shown,

$$E_{nq} \approx \text{constant} + \frac{\hbar^2 q^2}{2m_n^*}$$

just like the free particle (Equation 2.92) with $k \rightarrow q$.